Understanding Job Scam Victimization: A Behavioral Economics Perspective

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***Abstract*—Job scams represent an increasingly sophisticated form of cyber-enabled fraud, yet research has largely focused on their technical characteristics rather than the behavioral processes that make individuals vulnerable. This study advances a human-centered perspective by examining two dominant job-scam pathways, the pay-to-secure-employment model and the emerging task-based model, through the analytical lens of behavioral economics. Using anonymous survey data from university students, we assess how specific cognitive biases and heuristics, including the sunk cost fallacy, loss aversion, and social proof, shape susceptibility to these deceptive contexts. A mixed-methods approach integrating statistical modeling, sentiment analysis, and machine-learning classification is applied to identify psychological patterns associated with engagement and compliance. The findings reveal that vulnerability to job scams is not merely a consequence of informational deficits or technical deception but is systematically driven by predictable cognitive tendencies that influence judgment under uncertainty. These results demonstrate that traditional cybersecurity frameworks, which emphasize detection, awareness, or technological safeguards, insufficiently account for the behavioral mechanisms that scammers exploit. By empirically linking scam susceptibility to established constructs in behavioral economics, this study provides a conceptual bridge between the two fields and offers a scalable foundation for theory-informed intervention design. In doing so, it highlights the critical need for integrating behavioral-economic models into cybersecurity research to more fully explain and mitigate why individuals fall victim to employment fraud.**

***Index Terms*—Job scams, Behavioral economics, Cybercrime, Sunk cost fallacy, Loss aversion, Social proof, Cognitive biases**

I. Introduction

A. The Global Cybercrime Landscape

Cybercrime has become one of the most pervasive security and economic challenges of the digital era, inflicting substantial harm on individuals, businesses, and critical infrastructure worldwide. According to the Internet Crime Complaint Center (IC3) of the Federal Bureau of Investigation, more than 859,000 cybercrime complaints were recorded in 2024 alone, with reported losses exceeding $16.6 billion, an increase of 33% compared to the previous year [1]. Over the five-year period from 2020 to 2024, IC3 documented over 4.2 million complaints and $50.5 billion in cumulative losses, with daily complaint volumes rising from roughly 2,000 per month to nearly 2,000 per day [1]. While cryptocurrency investment fraud accounted for the largest financial losses and phishing and spoofing remained the most frequently reported crime categories, these figures collectively illustrate the breadth and escalating sophistication of cyber-enabled crime. Far from being a series of isolated incidents, cybercrime now constitutes a systemic threat that demands integrated behavioral, technological, and policy responses. Within this expanding landscape, one particularly insidious and rapidly growing subtype has gained prominence: employment fraud, commonly referred to as job fraud/scam.

B. The Rising Threat of Job Fraud

Within the broader cybercrime landscape, job fraud has emerged as one of the most rapidly expanding and financially consequential categories of online deception. Recent data from the U.S. Federal Trade Commission (FTC), one of the most comprehensive global sources for fraud reporting, shows that losses from job scams increased from $90 million in 2020 to $501 million in 2024, a more than five-fold rise, accompanied by a tripling in the number of reported cases [2]. In just the first six months of 2024, losses exceeded $220 million [2]. Although these figures represent the U.S. context, they are widely used as indicators of international cybercrime trends due to the FTC's rigorous reporting mechanisms. Moreover, prior research consistently demonstrates that fraud victimization is significantly underreported worldwide, often due to shame, embarrassment, or fear of judgment [3], [4]. As a result, job fraud constitutes a major yet often underestimated global cyber threat, with impacts extending well beyond the regions where data are systematically collected.

Contemporary job scams have adapted to digital platforms including WhatsApp, Telegram, encrypted messaging applications, and fraudulent job portals, leveraging the anonymity and global reach of the internet to target vulnerable populations with unprecedented efficiency. Modern employment fraud encompasses several distinct typologies: Pay-to-Get-a-Job scams, which require upfront payments for training materials, background checks, certification fees, or equipment purchases in exchange for promised employment opportunities that never materialize; Work-from-Home schemes that promise unrealistic income with minimal effort; Multi-Level Marketing (MLM) disguised as employment; and most alarmingly, Task-Based scams—a gamified variant that has driven much of the recent surge in job fraud losses.

Task-Based scams represent the most alarming trend, with reports increasing from zero in 2020 to 5,000 in 2023, then quadrupling to approximately 20,000 in just the first half of 2024, accounting for 38.8% of all job scam reports [2]. These scams typically begin with unsolicited messages via text, WhatsApp, or Telegram offering vague online work opportunities involving ”product boosting,” ”app optimization,” or content rating. Victims complete simple repetitive tasks and receive real payments initially, creating a false sense of legitimacy. The scam then pivots to an extraction phase, requiring victims to deposit their own money—often in cryptocurrency—to ”unlock” higher-level tasks and earnings that never materialize. Cryptocurrency losses to job scams hit $41 million in the first half of 2024, nearly double the $21 million reported in all of 2023 [2], highlighting how scammers exploit both psychological vulnerabilities and emerging payment technologies to facilitate untraceable transactions. This research focuses specifically on understanding the psychological mechanisms underlying two primary job fraud categories: Pay-to-Get-a-Job scams and Task-Based scams, examining how they exploit distinct cognitive vulnerabilities.

C. Limitations of Traditional Approaches

Traditional approaches to combating fraud have emphasized technological detection systems, demographic profiling of victims, legal prosecution of perpetrators, and awareness campaigns based on simple warnings [6], [7]. While these measures remain important, they fail to address a fundamental question: why do intelligent, educated individuals continue to engage with scams even when warning signs emerge? Scholarly research on job fraud specifically has been limited, with most attention focused on investment fraud [8], romance scams [9], advance-fee fraud [10], and online auction scams [6]. The existing job fraud literature has predominantly taken descriptive approaches documenting scam typologies and victim demographics or technical perspectives emphasizing detection algorithms [12], rather than systematically examining the underlying psychological decision-making processes that make these scams effective.

D. Behavioral Economics Framework

Recent advances in behavioral economics—the interdisciplinary field examining how psychological, cognitive, social, and emotional factors influence economic decision-making—provide powerful theoretical frameworks for understanding fraud victimization that have been underutilized in the job scam context. Behavioral economics challenges the traditional economic assumption of rational decision-making, demonstrating that humans systematically rely on mental shortcuts (heuristics), exhibit predictable biases in judgment, and make decisions influenced by emotional states, social context, and cognitive limitations [13]–[15]. This could explain why even financially sophisticated individuals fall victim to fraud—their knowledge is overridden by psychological mechanisms that scammers deliberately trigger.

For our scope, young adults represent a theoretically relevant population for investigating these mechanisms, as they face documented susceptibility factors including financial pressures [19], high unemployment rates [20], cognitive developmental factors related to risk assessment [17], [18], and increasing exposure to sophisticated online employment scams [2].

E. Research Objectives and Contributions

We look at how behavioral economic principles explain job fraud victimization among Indian university students, with specific attention to the psychological mechanisms exploited by Pay-to-Get-a-Job and Task-Based scam structures. Rather than testing predetermined hypotheses, we use an exploratory approach grounded in established behavioral economic theories to understand decision-making processes. We apply multiple theoretical frameworks including Sunk Cost Theory [14], [21], which explains why individuals continue investing in losing propositions to justify prior commitments; Loss Aversion from Prospect Theory [13], demonstrating that the psychological impact of losses exceeds equivalent gains; and Social Proof [22], showing how individuals rely on others' behaviors to guide their own decisions, particularly under uncertainty. To empirically investigate these mechanisms, we employ a mixed-methods approach integrating sentiment analysis to capture emotional dimensions of victim narratives, machine learning techniques to identify predictive patterns in victimization, and traditional statistical methods to test theoretical relationships.

We use our insights to suggest intervention design, that could inform the development of educational programs, warning systems, policy frameworks, and behavioral interventions that address the specific psychological vulnerabilities exploited by scammers rather than generic awareness campaigns.

II. Methodology

A. Research Design

This study employs a mixed-methods design within a cross-sectional framework, where we collect quantitative and qualitative data simultaneously via a single survey instrument using convenience and snowball sampling from the IIITH student community.

The mixed-methods approach combines structured quantitative scales (Likert scales, multiple-choice items) with open-ended qualitative responses, enabling statistical hypothesis testing while capturing contextual nuance. Quantitative components measure psychological constructs numerically and support statistical inference, while qualitative components reveal provide us with real insights not constrained by predetermined response categories.

B. Survey Instrument

1) Platform Selection: Google Forms was selected as the data collection platform based on the criteria presented in Table I.

2) Anonymous Design: The survey was designed to be completely anonymous to address the intense social stigma associated with fraud victimization. Table II presents the key reasons for this design choice.

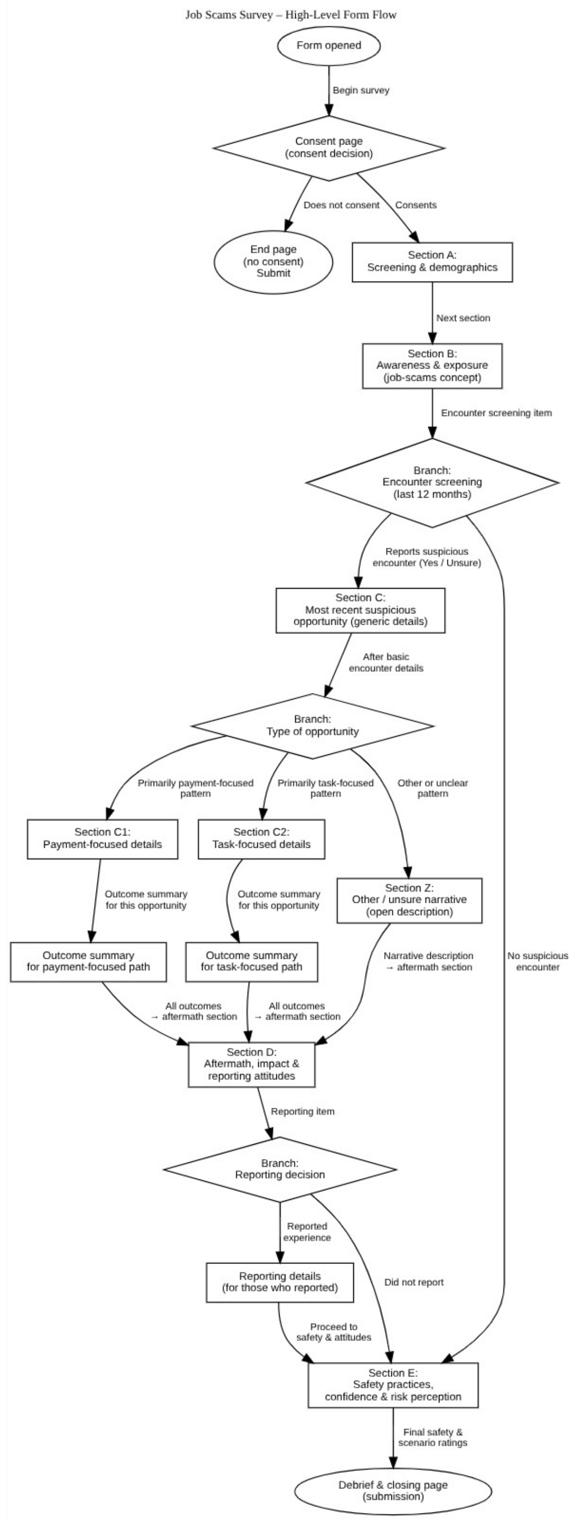

Fig. 1. Survey conditional flow diagram showing branching logic and section routing based on participant responses.

3) Target Population: The research targets the IIIT Hyderabad community and extended network, including students, faculty, staff, alumni, and their family members and friends.

C. Survey Structure and Sections

The survey comprises eight main sections with conditional branching based on participant responses (see Figure 1). Table III provides an overview of the survey architecture.

TABLE I
GOOGLE FORMS PLATFORM SELECTION RATIONALE

Criterion	Justification
Accessibility	Universal access without software installation
Familiarity	Student population regularly uses platform
Cost	Free for unlimited responses
Data Management	Automatic centralized storage with real-time updates
Privacy	No IP address or email collection possible
Conditional Logic	Branching based on responses reduces survey fatigue
Validation	Built-in rules for numeric fields and required items
Progress Tracking	Visual progress bar reduces abandonment

TABLE II
RATIONALE FOR ANONYMOUS SURVEY DESIGN

Factor	Benefit
Social Stigma	Reduces fear of judgment and self-recrimination
Response Honesty	Minimizes social desirability bias
Financial Disclosure	Enables specific payment amount reporting
Psychological Safety	Facilitates admission of cognitive biases
Response Rate	Higher participation for sensitive topics
Data Completeness	Reduces non-response to sensitive items

1) Section A: Screening and Demographics: Section A collects demographic and socioeconomic variables serving as control variables and enabling subgroup analyses. Table IV details the key demographic measures and their analytical purposes.

The income and household financial variables are particularly critical, as the same absolute monetary loss represents vastly different proportional harm across income levels, affecting loss aversion and sunk cost dynamics.

2) Section B: Awareness and Exposure: Section B measures baseline scam awareness and filters participants based on encounter experience. Table V summarizes the key measures.

The encounter screening question uses a 12-month recall window to balance recency with prevalence (ensuring sufficient victim sample size).

3) Section C: Most Recent Encounter: Section C documents the specific details of participants' most recent suspicious encounter. Table VI presents the core encounter variables.

The social proof and sunk cost influence scales help operationalize the key psychological constructs tested.

4) Type Router and Specialized Sections: Following the core encounter questions, participants categorize their experience using the type router question, which branches them to specialized sections. Table VII shows the routing logic.

a) Section C1: Payment-Based Scam Details: Section C1 collects detailed information about payment-based fraud encounters. Table VIII presents the key variables.

b) Section C2: Task-Based Scam Details: Section C2 collects parallel information for task-based encounters with additional gamification-specific measures. Table IX presents the key variables.

c) Section Z: Other/Unsure: Section Z provides a single comprehensive open-ended field for participants whose experiences may not fit the payment-based or task-based categories.

5) Section D: Aftermath and Reporting: Section D documents emotional impact and reporting behavior. Table X summarizes the key measures.

6) Section E: Safety, Attitudes, and Scenarios: Section E assesses protective behaviors, technical skills, and risk perception applicable to all participants regardless of victimization history. Table XI presents the key measures.

The three hypothetical scenarios test whether participants can recognize warning signs in abstract controlled contexts, enabling comparison between theoretical knowledge and practical application under real decision-making pressure.

D. Data Collection Procedure

The survey was distributed through multiple channels via WhatsApp groups, email lists, and social media.

No formal pilot testing with a separate sample was conducted.

E. Ethical Considerations

The research adhered to ethical standards for human subjects research involving vulnerable populations. Table XII summarizes the key ethical safeguards.

No compensation was provided to participants.

F. Data Analysis Techniques

The survey data was analyzed using various statistical methods to test the research hypotheses and understand behavioral patterns in job scam victimization. All analyses were conducted in Python using standard packages for statistics (SciPy, statsmodels), machine learning (scikit-learn), text analysis (transformers), and data visualization (matplotlib, seaborn, pandas).

1) Data Preparation:

a) Missing Data Handling: Missing data were not imputed to maintain data integrity. Each analysis used only complete cases for the specific variables involved, accepting reduced sample sizes over potentially biased estimates from imputation. We found this to be particularly critical with small datasets where artificial data points could substantially distort results.

b) Survey Flow Mapping: The conditional branching structure was mapped to understand differential response patterns across survey paths (Consent $\rightarrow$ Encounter Screening $\rightarrow$ Scam Type Router $\rightarrow$ Reporting). This mapping explained systematic missingness patterns where participants did not see questions inapplicable to their experiences.

2) Descriptive Statistical Analysis:

a) Frequency and Percentage Distributions: Table XIII presents the application domains for frequency analysis.

TABLE III
SURVEY SECTION OVERVIEW AND PURPOSE

Section	Purpose	Question Types	Routing Logic
Section 0: Consent	Obtain informed consent	Binary choice	Consent → A; No consent → End
Section A: Demographics	Collect demographic and socioeconomic variables	Multiple choice, checkboxes	All → B
Section B: Awareness	Measure scam awareness and exposure	Scales, binary choice	Yes/Unsure → C; No → E
Section C: Encounter Details	Document most recent suspicious encounter	Mixed: scales, checkboxes, text	Type-based routing
Section C1: Payment-Based	Detailed payment scam information	Numeric, scales, checkboxes, text	All → D
Section C2: Task-Based	Detailed task scam information	Numeric, scales, checkboxes, text	All → D
Section Z: Other/Unsure	Open narrative for ambiguous cases	Long-form text	All → D
Section D: Aftermath	Document emotional impact and reporting	Scales, binary choice, text	Report Yes → Details; No → E
Section E: Safety	Assess protective behaviors and risk perception	Checkboxes, scales	All → Debrief
Debrief	Provide resources and thank participants	Informational text	Submit

TABLE IV
SECTION A: DEMOGRAPHIC VARIABLES AND ANALYTICAL PURPOSE

Variable	Measurement	Analytical Purpose
Community Relationship	Single choice: Student, Faculty, Staff, Alumni, Family, Friend, Other	Population verification; subgroup comparison across relationship types
Age Band	Categorical ranges (Below 18, 18-20, 21-24, 25-29, 30-39, 40-49, 50+)	Control for developmental stage; test age-related vulnerability
Current Status	Multiple selection: Student, Employed, Preparing for jobs, Other	Identify job-seeking behavior as exposure factor
Job Search Channels	Multiple selection: 8 platform types (Campus, LinkedIn, WhatsApp, etc.)	Map scam exposure platforms; test platform-specific vulnerability
Geography	Country/region; State (if India)	Control for regional scam variation
Education Level	Ordered categories: High school to PhD	Test education as protective factor
Monthly Income	Binned ranges in INR	Critical for loss aversion and sunk cost analysis; contextualize financial harm
Household Financial Situation	5-point scale: Comfortable to In debt	Measure financial pressure as vulnerability factor

TABLE V
SECTION B: AWARENESS AND EXPOSURE MEASURES

Item	Format	Purpose
Job Scam Definition Box	Informational text with 8 defining characteristics	Ensure construct validity through common understanding
Prior Familiarity	1-5 scale (Not familiar - Very familiar)	Control variable; see awareness vs. susceptibility relationship
Encounter Screening	Binary + Unsure (Yes/Unsure/No)	Primary routing filter; distinguish victims from non-victims
Opportunity Sources	Multiple selection: 8 platform types	Cross-reference legitimate vs. scam platforms
Recognition Confidence	1-5 scale (Not confident - Very confident)	Self-efficacy measure; test for overconfidence effects

b) Central Tendency and Dispersion: For Likert scale variables, summary statistics were calculated including mean, median, standard deviation, and range. These measures characterized overall levels and variability for constructs including scam familiarity, recognition confidence, psychological influence scales, and risk perception ratings.

c) Subgroup Analysis: Analysis proceeded through hierarchical sample subsets based on conditional logic:

- Total sample: All valid survey responses
- Encounter subset: Participants reporting suspicious offers
- Payment subset: Participants who made financial payments
- Reporting subset: Participants who reported fraud to authorities

G. Hypotheses Testing

1) Sunk Cost Fallacy and Payment Escalation:

a) Theoretical Framework: The sunk cost fallacy describes the tendency for individuals to continue investing in a losing proposition to justify prior commitments of time, effort,

TABLE VI
SECTION C: CORE ENCOUNTER DOCUMENTATION VARIABLES

Variable	Measurement	Analytical Purpose
First Contact Platform	Single choice: 8 platform types	Test platform effects
Offer Type	Multiple selection: Internship, Full-time, Part-time, Remote task, Training, Other	Characterize scam bait tactics
Warning Signs Noticed	Multiple selection: 8 specific red flags + Other	Test awareness-action gap; identify overlooked vs. dismissed warnings
Engagement Level	Ordered categories: None, Brief (no payment), Including payment	Measure commitment escalation; primary behavioral outcome
Social Proof Influence	1-5 scale (Not at all - A lot)	**Test:** A measure of social proof susceptibility
Sunk Cost Influence	1-5 scale (Not at all - A lot)	**Test:** A measure of time/effort investment influence
Evidence Upload	Optional URL field	Triangulate self-report with artifacts

TABLE VII
TYPE ROUTER CATEGORIES AND ROUTING

Category Selection	Routes To
Payment-based scam: Upfront fees, refundable deposits, training charges	Section C1
Task-based scam: Paid micro-tasks with initial rewards, later investment requests	Section C2
Other/Unsure: Doesn't fit categories or uncertain	Section Z

or money [14], [21]. This cognitive bias violates the economic principle of rational decision-making, which dictates that only prospective costs and benefits should influence choices, not irrecoverable past investments. Arkes and Blumer [21] demonstrated that individuals who had invested more resources were significantly more likely to persist in failing courses of action, even when continuation was objectively disadvantageous. In the context of job scams, victims who have already invested time researching an opportunity, effort preparing application materials, or emotional energy anticipating employment may feel compelled to continue engagement—including making payments—to avoid the psychological discomfort of admitting their prior investments were wasted.

b) Hypotheses: **H1**: Higher self-reported sunk cost influence predicts greater likelihood of making payments to fraudulent job opportunities.

H1a: The relationship between sunk cost influence and payment behavior is moderated by financial vulnerability; the effect is stronger among individuals experiencing greater economic pressure.

c) Variable Operationalization: The sunk cost construct was directly measured using a single-item Likert scale positioned in Section C (Most Recent Encounter) of the survey instrument. The item asked: *"How much did the time and effort you had already invested influence your decision to continue?"* with responses on a 5-point scale ranging from 1 ("Not at all") to 5 ("A lot"). This direct self-report measure captures participants' conscious awareness of how prior investments influenced their subsequent decisions.

The primary dependent variable was payment behavior, operationalized as a binary indicator derived from the "Total amount paid (approx.)" item. Responses were coded as 1 (made payment) for any selection except "I did not pay" or missing data, and 0 (did not pay) otherwise.

For the moderation hypothesis (H1a), financial vulnerability was operationalized as a composite score integrating two dimensions from Section A demographics. The vulnerability index assigned 2 points for low income indicators (responses of "Less than Rs.10,000", "Rs.10k–20k", or "Not applicable"), plus 2 points for "Struggling (difficulty meeting expenses)" household status or 1 point for "In debt" status. This produced a 0-4 scale reflecting economic pressure, with higher scores indicating greater financial vulnerability.

d) Statistical Methods: Given the small sample size of participants who encountered suspicious job opportunities and the binary nature of the payment outcome, we employed the following methods.

Primary categorical analysis: We dichotomized the sunk cost influence variable at the median (creating high/low groups) and constructed 2×2 contingency tables crossing sunk cost level with payment status. Fisher's Exact Test [24] was applied rather than the chi-square test of independence, as Fisher's method provides exact p-values without relying on large-sample approximations. We calculated odds ratios to quantify the strength of association, along with risk ratios to express the multiplicative increase in payment probability.

Non-parametric comparison: Mann-Whitney U test compared sunk cost influence scores between those who made payments versus those who did not. Spearman's rank correlation coefficient assessed the monotonic relationship between sunk cost scores and payment behavior.

Bayesian bootstrap approach: To address uncertainty in effect size estimation with limited data, we implemented a bootstrap resampling procedure with 5,000 iterations. Each iteration randomly resampled the observed data with replacement, fit a logistic regression model predicting payment from sunk cost influence, and extracted the odds ratio. This generated a posterior distribution of plausible effect sizes, from which we calculated 95% credible intervals and the probability that the odds ratio exceeds 1 (indicating a positive effect). This Bayesian perspective provides probabilistic statements about effect existence that complement traditional null hypothesis testing.

TABLE VIII
SECTION C1: PAYMENT-BASED SCAM VARIABLES

Variable	Format	Purpose
First Payment Amount	Numeric (INR)	Quantify initial commitment anchor point
First Payment Reason	Paragraph text	Capture persuasion tactics and rationalizations
Total Payments Made	Numeric count	Measure escalation through repeated commitments
Total Amount Paid	Binned ranges (INR 500 to INR 50,000+)	Primary financial harm outcome; sensitive data protection
Decision Factors	Multiple selection: 10 influence factors	Deconstruct psychological mechanisms (urgency, authority, need)
Payment Methods	Multiple selection: UPI, Bank transfer, Cash, Wallet, Card, Other	Trace transaction
Refund/Guarantee Promises	1-5 scale (Never - Always)	Measure deceptive risk-reduction tactics
Persuasion Tactics Used	Multiple selection: 10 tactics	Map scammer influence methods
Outcome	Multiple choice: 5 outcome categories	Verify fraud vs. legitimate opportunity

TABLE IX
SECTION C2: TASK-BASED SCAM VARIABLES

Variable	Format	Purpose
Task Description	Paragraph text	Document task mechanics and operational details
Initial Small Payment	Binary (Yes/No/NA)	**Hypothesis:** Test reciprocity-based trust building
Average Reward Per Task	Numeric (INR)	Quantify economic bait amount
Gamification Elements	Multiple selection: Game mechanics (leaderboards, levels, streaks, etc.)	Identify behavioral activation tactics
Investment Request Timing	Binary (Yes/No/NA)	Capture bait-and-switch transition point
Tasks Completed	Numeric count	Behavioral measure of engagement depth
Trust from Small Payments	1-5 scale (Strongly disagree - Strongly agree)	**Hypothesis:** Measure reciprocity effect on trust
Fairness Perception	1-5 scale (Very unfair - Very fair)	Test exploitation normalization
Total Amount Paid	Binned ranges (identical to C1)	Enable cross-type financial harm comparison
Decision Factors	Multiple selection (identical to C1)	Enable cross-type psychological mechanism comparison
Payment Methods	Multiple selection (identical to C1)	Enable cross-type transaction method comparison
Outcome	Multiple choice: 6 outcome categories	Verify fraud vs. legitimate opportunity

TABLE X
SECTION D: AFTERMATH AND REPORTING VARIABLES

Variable	Format	Purpose
Barriers to Stopping	Paragraph text	Qualitative validation of psychological trap mechanisms
Emotional Impact: Anger	1-5 scale (Not at all - Extremely)	Measure discrete emotion
Emotional Impact: Anxiety	1-5 scale (Not at all - Extremely)	Assess mental health
Emotional Impact: Embarrassment	1-5 scale (Not at all - Extremely)	Quantify stigma requiring de-shaming campaigns
Reported Anywhere	Binary (Yes/No)	Primary reporting behavior outcome
Where Reported	Multiple selection: 7 formal/informal channels	Map help-seeking pathways
Reporting Outcome	Paragraph text (optional)	Document system responsiveness

TABLE XI
SECTION E: SAFETY AND RISK PERCEPTION MEASURES

Variable	Format	Purpose
Reporting Knowledge	1-5 agreement scale	Test whether knowledge barriers prevent reporting
Protective Checks Performed	Multiple selection: 7 verification behaviors	Document actual protective practices
Technical Skill: URL Checking	1-5 comfort scale	Measure domain verification competence
Technical Skill: Impersonation Detection	1-5 comfort scale	Measure visual spoofing detection competence
Risk Scenario 1: Task-based	1-5 risk scale	Test recognition of immediate-reward warning signs
Risk Scenario 2: Payment-based	1-5 risk scale	Test recognition of refundable-fee tactics
Risk Scenario 3: Social Proof	1-5 risk scale	Test resistance to crowd validation signals

TABLE XII
ETHICAL SAFEGUARDS AND PROTECTIONS

Protection	Implementation
Informed Consent	Mandatory consent page with full disclosure
Anonymity	No PII collection; extensive warnings
Compensation	None; voluntary participation
Debrief	Resources and reporting information provided

TABLE XIII
APPLICATIONS OF FREQUENCY DISTRIBUTION ANALYSIS

Analysis Domain	Variables Examined
Sample Characterization	Consent rates, encounter rates, demographics (age, education, employment, geography)
Behavioral Patterns	Job search platforms, verification behaviors, engagement levels

Logistic regression: We fit binary logistic regression models with payment as the outcome and sunk cost influence as the primary predictor. Model 1 included sunk cost alone; Model 2 added vulnerability score; Model 3 included an interaction term (sunk cost × vulnerability) to test the moderation hypothesis. Given the small sample relative to the number of parameters, we applied penalized regression with L2 regularization to reduce overfitting risk.

Propensity score methods: To control for potential confounding variables (social proof influence, vulnerability, familiarity with scams), we calculated propensity scores representing each participant's probability of reporting high sunk cost influence. Participants with high sunk cost were matched to those with low sunk cost having similar propensity scores, creating balanced comparison groups. We then compared payment rates between matched pairs using McNemar's test for paired binary data.

Sensitivity analysis: We tested whether results depended on arbitrary threshold choices by varying the cutpoint for defining "high" sunk cost ($\geq$2.5, $\geq$3, $\geq$3.5, $\geq$4) and re-running Fisher's Exact Test at each threshold. Consistent findings across thresholds would indicate robust effects independent of categorization decisions.

2) Loss Aversion and Fear-Driven Decision Making:

a) Theoretical Framework: Loss aversion, [13], posits that individuals experience losses as psychologically more impactful than equivalent gains, with empirical estimates from subsequent research suggesting loss aversion coefficients typically slightly over 2 [26]. This asymmetry in value perception leads to risk-seeking behavior in the domain of losses as individuals take desperate actions to avoid crystallizing losses. In the job scam context, loss aversion manifests through fear of missing out on employment opportunities, particularly when individuals perceive themselves as having limited alternatives. When scammers create artificial scarcity ("only 5 spots remaining"), impose time pressure ("offer expires in 24 hours"), or exploit genuine financial desperation ("you need this income"), they activate loss aversion mechanisms that override rational skepticism. Unlike the sunk cost fallacy, which is backward-looking (justifying past investments), loss aversion is forward-looking, driven by anticipated regret over missing potential opportunities [14].

b) Hypotheses: **H2**: Self-reported fear of losing opportunity predicts higher likelihood of making payments to fraudulent job opportunities.

H3: Self-reported urgency and time pressure predict higher likelihood of making payments to fraudulent job opportunities.

H4: Financial vulnerability amplifies loss aversion effects, such that the relationship between fear of loss and payment behavior is stronger among individuals experiencing greater economic pressure.

c) Variable Operationalization: Loss aversion constructs were measured through multiple-selection checklist items in Section C1 (Payment-Based Scam Details) and Section C2 (Task-Based Scam Details), positioned under the question: *"What affected you to pay at the time? (only if applied, select all top reasons)."*

Three binary indicators captured distinct facets of loss aversion:

- **Fear of losing opportunity**: Binary indicator (1 = selected, 0 = not selected) for the checkbox option "Fear of losing opportunity."

- **Urgency/time pressure**: Binary indicator for "Sense of urgency or time pressure influence."
- **Need for income**: Binary indicator for "Need for income," representing financial desperation as an extreme form of loss aversion.

A composite **loss aversion score** was constructed by summing the fear of opportunity and urgency indicators, producing a 0-2 scale. This composite captured the intensity of loss-driven motivation, with higher scores indicating multiple simultaneous loss aversion mechanisms.

The dependent variable remained the binary payment indicator (made payment vs. did not pay), and the moderating variable remained the 0-4 financial vulnerability composite score described earlier.

d) Statistical Methods: **Fisher's Exact Test**: We constructed 2×2 contingency tables crossing each loss aversion indicator (fear of opportunity: yes/no; urgency pressure: yes/no) with payment status. Odds ratios quantified the association strength, and Cohen's h measured standardized effect sizes for proportional differences.

Risk ratio calculation: For each loss aversion indicator, we calculated the risk ratio comparing payment rates between those who endorsed the indicator versus those who did not.

Stratified analysis: We divided the sample into high vulnerability (score $\geq$ median) and low vulnerability (score $<$ median) groups, then separately examined the relationship between fear of loss and payment within each stratum. Stronger associations in the high vulnerability group would support the moderation hypothesis.

Logistic regression models: We fit three hierarchical logistic regression models predicting payment:

- **Model 1**: Fear of opportunity + urgency pressure + sunk cost influence (comparing psychological mechanisms)
- **Model 2**: Model 1 + vulnerability score (testing main effect of vulnerability)
- **Model 3**: Model 2 + vulnerability × fear interaction term (testing moderation)

Model comparisons assessed whether adding vulnerability and interaction terms improved predictive accuracy. Given the small sample relative to the number of parameters, we used the BFGS optimization algorithm with maximum 100 iterations and applied regularization where convergence was problematic.

Comparison with sunk cost: To determine whether loss aversion or sunk cost fallacy represented the dominant psychological mechanism, we compared effect sizes (odds ratios, risk ratios) across the two frameworks. We also examined whether loss aversion indicators and sunk cost influence were correlated, which would suggest overlapping psychological processes versus independent pathways to payment.

3) Social Proof and Informational Cascades:

a) Theoretical Framework: Social proof, systematically analyzed by Cialdini [22], describes the psychological phenomenon where individuals look to others' behaviors to guide their own decisions, particularly under conditions of uncertainty. This heuristic operates on the assumption that if many others are engaging in a behavior, it must be correct or safe. Bikhchandani et al. [27] formalized this as informational cascades, wherein individuals rationally infer information from others' actions, leading to herding behavior that can persist even when based on incorrect initial signals. In job scam contexts, fraudsters exploit social proof through fabricated testimonials, inflated follower counts, fake reviews, professional-looking websites with "verified" badges, and claims of widespread participation. These cues create the illusion of legitimacy and social validation, lowering skepticism.

b) Hypotheses: **H5**: Higher self-reported social proof influence predicts greater likelihood of making payments to fraudulent job opportunities.

H5a: The relationship between social proof influence and payment behavior is moderated by age, such that the effect is stronger among younger participants.

H5b: Professional appearance elements and social proof elements create a multiplicative effect on payment likelihood.

c) Variable Operationalization: The social proof construct was measured using a single-item Likert scale in Section C (Most Recent Encounter): *"How much did social proof (likes/testimonials) influence your decision to continue?"* with responses on a 5-point scale from 1 ("Not at all") to 5 ("A lot"). The item included extensive help text defining social proof as: *"professional-looking websites, verified-looking profiles, high numbers of likes/followers, testimonials from supposed previous participants, positive reviews, celebrity/influencer endorsements, or any indication that many others have trusted this opportunity."* to ensure participants understood the full scope of social proof tactics.

For the age moderation hypothesis (H5a), age was operationalized using the midpoint values of categorical age bands from Section A demographics (e.g., 18–20 $\rightarrow$ 19 years, 21–24 $\rightarrow$ 22.5 years). A binary age grouping divided participants into younger ($\leq$24 years) versus older ($>$24 years).

For the multiplicative interaction hypothesis (H5b), two binary indicators were extracted from the Section C checklist item *"Which elements did you notice? (select all that apply)"*:

- **Professional appearance**: Binary indicator (1 = selected, 0 = not selected) combining responses mentioning professional branding, websites, verified identifiers, or official-looking materials.
- **Social proof elements**: Binary indicator for responses mentioning testimonials, likes, views, social proof, or crowd validation signals.

The dependent variable remained the binary payment indicator.

d) Statistical Methods: **Mann-Whitney U test**: We compared social proof influence scores (1-5 Likert scale) between participants who made payments versus those who did not.

Spearman rank correlation: We calculated Spearman's ρ to assess the monotonic relationship between social proof scores and payment behavior, both overall and separately within younger and older age groups. Comparing correlation magnitudes across age strata tests the moderation hypothesis (H5a).

Multiplicative interaction analysis: To test whether professional appearance and social proof elements combine synergistically (H5b), we constructed a 2×2 contingency table crossing professional appearance (present/absent) by social proof elements (present/absent) and examined payment rates in each cell. We calculated the expected additive effect as: Payment rate(professional only) + Payment rate(social proof only) - Payment rate(neither). If the observed payment rate when both elements are present exceeds this additive expectation, it suggests a multiplicative interaction.

Mediation analysis: We explored whether social proof operates through trust mechanisms. The hypothesized pathway was: Social proof influence → Trust susceptibility → Payment decision. We operationalized trust susceptibility as the inverse of recognition confidence (6 - confidence scale score), where lower confidence indicates higher susceptibility to deceptive trust signals. The mediation procedure tested: (a) whether social proof predicts trust susceptibility, (b) whether trust susceptibility predicts payment controlling for social proof, and (c) whether the direct effect of social proof on payment diminishes when controlling for trust. We used ordinary least squares regression for the continuous mediator and logistic regression for the binary outcome. Bootstrap resampling with 1,000 iterations generated 95% confidence intervals for the indirect effect (a × b path).

Random Forest feature importance: To compare the relative predictive importance of social proof versus other psychological mechanisms (sunk cost, loss aversion, vulnerability), we trained a Random Forest classifier predicting payment status from all behavioral and demographic features. We used 1,000 trees with maximum depth of 3 and minimum leaf size of 3 to prevent overfitting with small samples. Cross-validated ROC-AUC scores assessed overall model discrimination.

Logistic regression: We fit logistic regression models including social proof influence as a predictor alongside sunk cost and vulnerability, allowing comparison of standardized coefficients to assess relative importance of different psychological mechanisms.

4) Emotional Response Analysis via Natural Language Processing:

a) Theoretical Framework: Written narratives about traumatic or significant experiences contain emotional signatures that may predict subsequent behaviors and outcomes. In the fraud victimization context, the emotional tone of victim narratives may reveal psychological processes beyond what self-report scales capture, including retrospective emotional regulation, cognitive reappraisal of the experience, and ongoing psychological impact.

Unlike the hypothesis-driven analyses for sunk cost, loss aversion, and social proof, this component employed an exploratory approach to investigate whether emotions extracted from free-text narratives relate to behavioral outcomes.

b) Research Questions: **RQ1**: What emotions dominate victim narratives about job scam experiences?

RQ2: Does emotional tone extracted from written narratives predict payment amounts?

RQ3: Does emotional tone extracted from written narratives predict reporting behavior?

RQ4: Do emotion distributions differ systematically across scam types (payment-based vs. task-based vs. other)?

c) Variable Operationalization: **Text data source**: We aggregated free-text responses from four optional narrative fields distributed across the survey:

- Section C1: “Share Your Experience (Optional)” [payment-based path]
- Section C2: “Share Your Experience (Optional)” [task-based path]
- Section Z: “Share Your Experience (Other/Unsure)”
- Section D: “When you realized it might be suspicious or fraudulent, what stopped you from stopping earlier?”

Responses from these fields were concatenated into a single combined narrative per participant. Only participants who provided at least 10 characters of text were included in the sentiment analysis subsample, as very brief responses (e.g., single words) lack sufficient context for reliable emotion classification.

Sentiment classification: We applied the pre-trained DistilBERT sentiment analysis model. This model classifies text into binary sentiment categories (POSITIVE or NEGATIVE) and provides a confidence score (0-1) for the assigned label. Text inputs were truncated to 512 tokens (the model’s maximum sequence length).

Emotion classification: We applied the pre-trained DistilRoBERTa emotion classification model, which categorizes text into six discrete emotions: anger, fear, joy, sadness, surprise, and neutral. The model outputs both a categorical emotion label and a confidence score (0-1) for the assigned emotion.

Outcome variables:

- **Payment amount**: Numeric midpoint values derived from the categorical “Total amount paid (approx.)” responses, providing a continuous measure of financial harm.
- **Reporting behavior**: Binary indicator from the “Did you report this experience anywhere?” item (Yes = 1, No = 0).
- **Scam type**: Categorical variable determined by the type router question: payment-based, task-based, or other/unsure.

d) Statistical Methods: **Descriptive analysis**: We calculated frequency distributions and percentages for each emotion category in the overall sample and stratified by scam type. This characterized the emotional landscape of victim narratives.

Kruskal-Wallis H-test: To test whether payment amounts differed across emotion categories (RQ2), we applied the Kruskal-Wallis test, a non-parametric alternative to one-way ANOVA appropriate for comparing continuous outcomes across multiple independent groups without assuming normality. The test assesses whether the distribution of payment amounts is identical across emotion categories. For post-hoc pairwise comparisons, we used Mann-Whitney U tests with false discovery rate (FDR) correction for multiple comparisons.

Chi-square test of independence: To examine whether emotion categories predicted reporting behavior (RQ3), we constructed a contingency table crossing emotion label (6 categories) by reporting status (reported vs. not reported). The chi-square test assessed whether the proportion of reporters differed significantly across emotion categories. Fisher's Exact Test was substituted when expected cell frequencies fell below 5.

Cross-tabulation by scam type: To address RQ4, we created contingency tables showing the distribution of emotions within each scam type, calculating row percentages (percentage of each scam type expressing each emotion). This descriptive analysis identified whether certain emotions were more prevalent in particular scam contexts (e.g., fear in task-based scams, anger in payment-based scams).

e) Methodological Considerations and Limitations: Several important limitations qualify the sentiment analysis findings. First, the emotion classification models could be trained on general English text corpora (tweets, movie reviews, product feedback) rather than victim narratives specifically, potentially reducing classification accuracy in this specialized domain. Second, temporal ambiguity exists regarding whether extracted emotions reflect feelings during the scam experience, during the writing of the narrative, or about the scam in retrospect—distinctions that cannot be disentangled from cross-sectional text data. Third, retrospective narratives may exhibit emotion regulation or cognitive reappraisal, such that expressed emotions differ from in-the-moment experiences. Fourth, the subsample with meaningful text responses was substantially smaller than the full survey sample, limiting statistical power and potentially introducing selection bias.

5) Missing Data as Behavioral Signal:

a) Theoretical Framework: Traditional missing data frameworks distinguish between three mechanisms [34]: Missing Completely At Random (MCAR), where missingness is unrelated to any variables; Missing At Random (MAR), where missingness depends on observed variables; and Missing Not At Random (MNAR), where missingness depends on the unobserved values themselves. In survey research on sensitive topics, missing data may not represent random noise but rather systematic behavioral signals reflecting psychological avoidance, privacy concerns, or cognitive resource depletion.

Several behavioral economic theories predict non-random missingness patterns in fraud victimization surveys. The **Ostrich Effect** [35] describes the tendency to avoid negative financial information, wherein investors check portfolios less frequently during market downturns to avoid confronting losses. Applied to survey disclosure, victims may avoid reporting payment amounts to prevent psychological salience of financial losses. **Self-Concept Protection** theory, derived from cognitive dissonance [36], suggests that admitting embarrassment or acknowledging emotional impact threatens self-image as competent and rational, motivating selective non-disclosure of emotions while potentially disclosing less threatening factual information. Ego depletion [39] could predict that cognitive resources for survey completion are limited and depletable, leading to increased item non-response as mental fatigue accumulates through lengthy questionnaires.

b) Research Questions: **RQ5**: Do victims systematically avoid disclosing financial losses, consistent with loss aversion operating at the reporting stage itself?

RQ6: Do victims selectively avoid disclosing embarrassment relative to other emotions (anger, anxiety)?

RQ7: Does non-reporting of fraud reflect cost calculations where the costs of reporting exceed expected benefits?

RQ8: Does missingness increase systematically with question position in the survey, consistent with cognitive fatigue effects?

RQ9: Are financial non-disclosure and emotional non-disclosure correlated, suggesting a dual avoidance strategy?

c) Variable Operationalization: **Missingness indicators**: Binary variables (0 = data present, 1 = missing) were created for key sensitive items:

- **Financial disclosure**: Missing status on "Total amount paid (approx.)" among payment-based scam respondents who were routed to this question.
- **Emotional disclosure**: Missing status on three emotion scales from Section D: "After this experience, how much did you feel: Anger," "Anxiety," and "Embarrassment."
- **Income disclosure**: Missing status or selection of "Prefer not to say" on "Current monthly personal income (approx.)" in Section A.
- **Reporting behavior**: Not actually missing—participants answered "Yes" or "No" to whether they reported the fraud—but the 91.9% "No" response rate was interpreted as revealed preference regarding transaction costs.

Predictors of missingness:

- **Loss magnitude**: Categorical grouping of payment amounts (No loss, Low, Medium, High) for those who disclosed amounts.
- **Vulnerability score**: The 0-4 composite financial pressure index.
- **Question position**: Sequential ordering of questions in the survey instrument (1 through 62), derived from the survey flow structure.
- **Scam encounter status**: Binary indicator distinguishing participants who encountered scams from those who did not.

d) Statistical Methods: **Descriptive analysis**: We calculated missingness rates (percentages) for each variable overall and stratified by relevant subgroups (e.g., scam type, vulnerability level, payment status). This characterized the scope and distribution of non-response across the dataset.

Chi-square tests of independence: We constructed contingency tables examining relationships between categorical predictors and missingness patterns:

- Loss magnitude category × Emotional data disclosure (testing whether larger losses predict emotional non-disclosure)

- Income disclosure status × Scam encounter status (testing whether victims differentially withhold income information)
- Reporting behavior × Emotional disclosure (testing whether non-reporters also avoid emotional disclosure)

Spearman rank correlation: To test the cognitive fatigue hypothesis (RQ8), we calculated Spearman's ρ between question position (1-62) and missingness percentage across all questions. A significant positive correlation would indicate that missingness increases as the survey progresses, consistent with ego depletion. This analysis treated each question as a data point, with position as the independent variable and missingness rate as the dependent variable.

Missingness correlation matrix: We computed pairwise Pearson correlations among binary missingness indicators for the key sensitive items (payment amount, anger, anxiety, embarrassment, income). High positive correlations indicate that participants who skip one item tend to skip others, suggesting categorical avoidance rather than item-specific non-response. We visualized this correlation structure using a heatmap to identify clustering patterns.

Stratified analysis: We examined emotional missingness rates separately within vulnerable versus non-vulnerable subgroups, and within payment-based versus task-based scam experiences.

e) Methodological Considerations: A critical complication in this analysis is the survey's conditional branching logic, which creates structural missingness by design. Participants who did not encounter scams were intentionally routed past all scam-specific questions, generating missingness that reflects appropriate survey flow rather than psychological avoidance. Similarly, participants categorizing their experience as payment-based never saw task-based questions and vice versa. This structural missingness must be carefully distinguished from behavioral non-response.

To separate these mechanisms, we calculated missingness rates only within samples where questions were applicable. For example, emotional disclosure rates were computed only among the subsample who encountered scams (n=37), as non-encounters were structurally prevented from seeing Section D. Similarly, payment amount missingness was analyzed only among payment-based scam respondents who were routed to Section C1.

III. Results

A. Descriptive Statistics

1) Sample Characteristics and Demographics: The survey collected 96 responses between October 7 and November 1, 2025, over a 25-day period. Of these, 91 participants (94.8%) provided informed consent and completed the survey. The respondent pool consisted primarily of current students (84.5%), with smaller representation from staff members (8.3%) and alumni (3.6%). The sample skewed young, with 63.7% aged 18–20 years and 23.1% aged 21–24 years, reflecting the university's undergraduate-heavy population. Geographically, 96.7% of respondents resided in India, with notable concentration in Telangana state (28.6% of Indian respondents). Figure 2 presents the demographic composition of the sample.

52.7% had completed high school or secondary education, while 25.3% held bachelor's degrees. Employment status aligned with this profile, as 41.8% identified as students actively preparing for jobs or internships, 40.7% were students not currently job-seeking, and only 9.9% reported full- or part-time employment.

a) Financial Context.: The sample's financial profile revealed significant economic vulnerability among certain subgroups. Monthly personal income data showed that 64.8% reported no income (unemployed students), while 14.3% earned less than Rs. 10,000 per month. When asked about household financial situations, 49.5% described their households as comfortable (meeting expenses with savings capacity), 29.7% as manageable (meeting expenses without savings), 6.6% as struggling to meet expenses, and 5.5% as being in debt. An additional 8.8% preferred not to disclose their financial situation. Figure 3 displays the distribution of income and household financial situations.

2) Scam Awareness and Self-Assessed Competencies:

a) Prior Knowledge and Confidence.: Participants reported moderate familiarity with the concept of job scams (Mean = 3.38, Median = 3.0 on a 5-point scale), with 30.8% rating their familiarity as 4 and an equal percentage rating it as 3. Only 19.8% reported low familiarity (ratings of 1–2). Self-assessed confidence in recognizing job scams when encountered showed a similar distribution (Mean = 3.33, Median = 3.0), though slightly more respondents clustered at lower confidence levels: 23.1% rated their confidence as 1–2, while 47.3% rated it as 4–5. Figure 4 presents the distributions of familiarity and confidence ratings alongside other awareness metrics.

These measures reveal a sample with moderate self-perceived preparedness but also substantial variance. Approximately one-quarter of respondents lacked confidence in their scam recognition abilities despite general awareness of the phenomenon.

b) Reporting Knowledge and Technical Skills.: In stark contrast to general scam awareness, knowledge of reporting mechanisms was notably deficient. When asked about familiarity with where and how to report suspicious offers, participants provided a mean rating of only 2.47 (Median = 2.0), with 58.4% rating their knowledge as 1–2 (low). This represents a critical knowledge gap: while most respondents could conceptually recognize scam risks, fewer than half knew the procedural steps to report encounters.

Technical skills related to scam detection showed moderate competency. Comfort with checking URLs and domain authenticity averaged 3.21 out of 5 (Median = 3.0), with 45.1% rating themselves 4–5. Comfort with spotting impersonation or lookalike pages averaged 3.00 (Median = 3.0), with a more even distribution across rating levels and 34.1% rating themselves 4–5. These self-assessments suggest reasonable but not exceptional technical literacy for digital fraud detection.

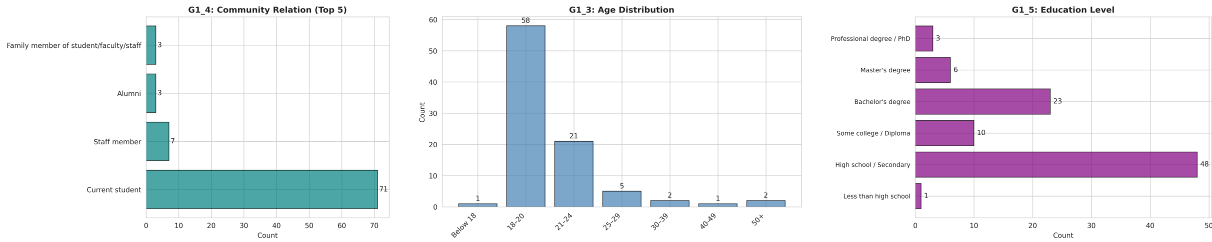

Fig. 2. Sample demographics: (a) Community relation showing student dominance, (b) Age distribution concentrated in 18–24 range, (c) Education level with majority at high school/secondary.

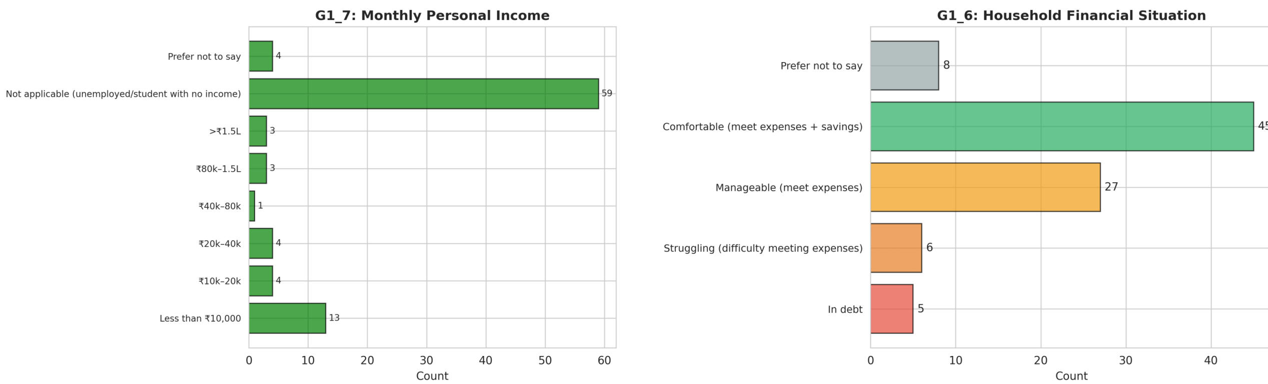

Fig. 3. Financial context: (a) Monthly personal income showing majority with no income, (b) Household financial situation with nearly half reporting comfortable circumstances but notable vulnerable populations (6.6% struggling, 5.5% in debt).

c) Verification Behaviors.: The most common verification strategy involved checking company legitimacy through official channels (LinkedIn, company websites) at 75.8%, followed by a general heuristic to avoid upfront payments (61.5%). Domain or email authenticity checks were performed by 54.9% of respondents, while 51.6% consulted friends, mentors, or family for advice. However, more technical verification methods such as reverse-image searches for visual fraud detection were employed by only 18.7% of respondents. 96.7% reported performing at least one regular verification check, suggesting good engagement with some form of due diligence.

d) Risk Perception.: All three scenarios received high mean risk ratings, indicating general sensitivity to common fraud patterns (see Figure 5):

- Scenario 1 (remote task with immediate payment): Mean = 3.84, Median = 4.0
- Scenario 2 (refundable training kit fee): Mean = 4.16, Median = 4.0
- Scenario 3 (high social proof, requests PAN/Aadhaar documents): Mean = 4.29, Median = 5.0

The highest risk ratings were assigned to Scenario 3, which combined social proof elements with requests for sensitive government identification documents. This suggests that respondents recognized the compounding risk of multiple fraud indicators. However, high risk perception did not necessarily translate to avoidance behavior.

3) Scam Encounter Characteristics: Of the 91 consenting participants, 37 completed the encounter-specific questions after indicating they had encountered suspicious job or internship offers in the past 12 months (38.5% encounter rate among those who reached the screening question).

a) Scam Typology and Channels.: Among the 37 encounters, scam types were distributed as follows: 40.5% were payment-based scams requiring upfront fees for job access, application processing, or training materials (15/37); 32.4% were task-based scams offering paid micro-tasks such as product rating or app downloads (12/37); and 27.0% did not clearly fit either category or respondents were unsure (10/37).

The primary contact channels through which these opportunities reached respondents are shown in Figure 6. WhatsApp and Telegram dominated as initial contact methods (51.4%), followed distantly by email (16.2%), LinkedIn (10.8%), and Instagram/Facebook (8.1%). Regarding what was offered, internships were most common (62.2%), followed by part-time positions (35.1%), remote tasks or gigs (29.7%), and training or certification programs tied to employment (27.0%). Respondents could select multiple categories if applicable.

b) Scam Elements and Tactics.: Respondents identified various suspicious elements they noticed in the offers (multiple selections allowed). As shown in Figure 7, the most frequently observed tactic was pressure or urgency to act immediately, reported in 54.1% of encounters (20/37). This was closely followed by requests for upfront payment for application,

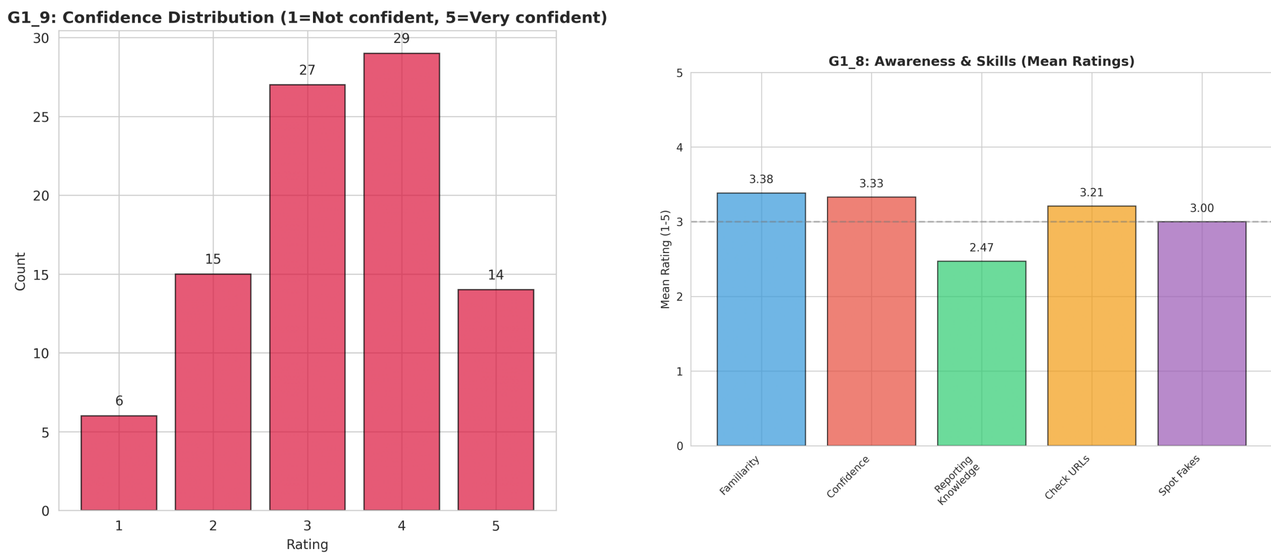

Fig. 4. Scam awareness and skills: (a) Confidence in recognizing scams (Mean=3.33), (b) Awareness skills showing moderate technical competencies (URL checking Mean=3.21, Spotting fakes Mean=3.00).

processing, or training (43.2%, 16/37) and promises of "small easy tasks" with quick rewards (43.2%, 16/37). High pay offers with minimal qualification requirements appeared in 35.1% of encounters (13/37). Social proof elements such as testimonials, likes, or views were noted in 10.8% of cases (4/37).

Encounters typically featured multiple suspicious elements simultaneously. The mean number of elements noticed per encounter was 2.2 (range 1–5), with 35.1% of encounters presenting a single element, 24.3% presenting two elements, 29.7% presenting three elements, and 10.8% presenting four or more elements.

4) Platform-Specific Engagement Patterns: Engagement rates varied substantially across initial contact platforms, as illustrated in Figure 8. Among encounters originating on Instagram or Facebook, 100% resulted in some level of engagement (3/3 encounters), though the small sample size limits generalizability. LinkedIn-initiated encounters showed 75.0% engagement (3/4), while phone or SMS contacts yielded 50.0% engagement (1/2). Email encounters resulted in 33.3% engagement (2/6), and WhatsApp/Telegram – despite being the most common contact channel – exhibited the lowest engagement rate at 26.3% (5/19).

5) Scam Outcomes: Among the 15 respondents who encountered payment-based scams and provided outcome information, 46.7% selected "Other" as the outcome, 40.0% reported they did not receive what was promised and suspected fraud, 6.7% indicated the situation was still ongoing or they were waiting, and 6.7% received partial benefits but not as promised.

For the 12 task-based scam encounters with outcome data, 41.7% selected "Other," 25.0% reported being asked to pay money but not receiving promised rewards, 16.7% received some rewards but not all as promised, and 16.7% did not receive promised rewards and suspected fraud.

These outcome distributions reveal that explicit fraud (no delivery of promises) was recognized in 40.0% of payment-based cases and 41.7% of task-based cases when combining the "suspected fraud" and "asked to pay/no rewards" categories. However, the high proportion of "Other" responses (46.7% and 41.7% respectively) suggests either diverse outcome trajectories not captured by preset categories or ambiguity in how respondents classified their experiences.

B. Inferential Statistics

C. Hypothesis Testing

1) Sunk Cost Fallacy and Payment Escalation: We tested whether self-reported sunk cost influence—the extent to which already-invested time and effort affected decisions to continue engagement—predicted payment behavior among the 37 participants who encountered suspicious job opportunities. Of these, 16 (43.2%) reported making payments.

a) Primary Categorical Analysis (H1).: Table XIV presents the 2×2 contingency table crossing sunk cost level with payment status.

TABLE XIV
PAYMENT BEHAVIOR BY SUNK COST INFLUENCE LEVEL (N=37)

Sunk Cost Influence	**No Payment**	**Made Payment**	**Total**
Low (1-2)	15 (68.2%)	7 (31.8%)	22
High (3-5)	6 (40.0%)	9 (60.0%)	15
Total	21 (56.8%)	16 (43.2%)	37

The high sunk cost group exhibited a payment rate of 60.0% (9/15) compared to 31.8% (7/22) in the low sunk cost group, yielding a risk ratio of 1.89. Fisher's Exact Test, indicated a non-significant association ($p = 0.107$, two-tailed). While this p-value exceeds the conventional $\alpha = 0.05$ threshold, the

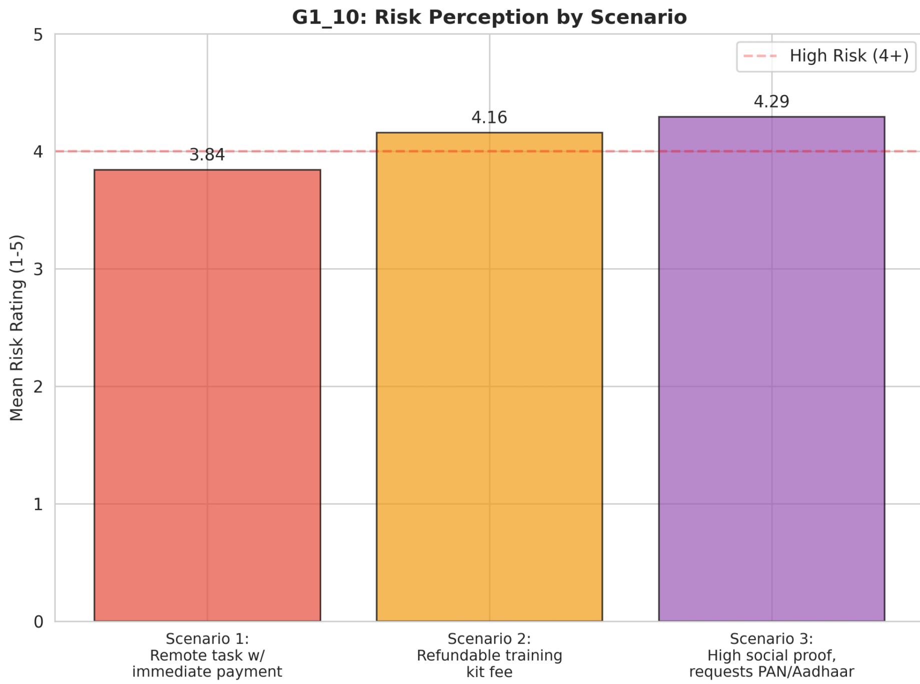

Fig. 5. Risk perception across three hypothetical scam scenarios. All scenarios rated as high risk (means $\geq$3.84), with Scenario 3 (social proof + sensitive documents) receiving highest ratings (Mean=4.29).

effect size measures suggest practical importance: the odds ratio of 3.21 indicates that participants reporting high sunk cost influence had more than three times the odds of making payments compared to those reporting low influence. Cramér's V = 0.224 represents a small-to-medium effect size.

b) Bayesian Credible Interval Estimation.: Figure 9 panel (a) displays the posterior distribution.

The posterior distribution yielded a median odds ratio of 1.60 (Mean = 1.80), with a 95% credible interval spanning [0.82, 3.80]. 92.3% of the posterior mass fell above OR = 1, indicating a 92.3% probability that sunk cost influence increases payment odds when incorporating sampling uncertainty. This Bayesian perspective suggests substantial evidence for a positive effect despite the classical test's non-significance.

c) Non-Parametric Rank-Based Tests.: Mann-Whitney U test compared sunk cost influence scores (treating the 1-5 scale as ordinal) between payers and non-payers. The test yielded U = 125.5, p = 0.180, with a rank-biserial correlation of 0.25, indicating that payers tended to report higher sunk cost influence than non-payers, though not at statistically significant levels. The common language effect size of 0.37 indicates a 37.4% probability that a randomly selected payer would have a higher sunk cost score than a randomly selected non-payer, compared to the 50% expected under no effect.

Spearman's rank correlation between sunk cost scores and payment behavior was ρ = 0.224 (p = 0.185), again suggesting a positive but non-significant monotonic relationship.

d) Moderation by Financial Vulnerability (H1a).: We tested whether the sunk cost-payment relationship was stronger among financially vulnerable individuals by examining payment rates stratified by both sunk cost level and vulnerability status (Table XV).

TABLE XV
PAYMENT RATES BY SUNK COST INFLUENCE AND FINANCIAL VULNERABILITY

	Low Vulnerability (score 0-1)		**High Vulnerability (score 2-4)**	
Sunk Cost	**n**	**Payment Rate**	**n**	**Payment Rate**
Low (1-2)	18	27.8% (5/18)	4	50.0% (2/4)
High (3-5)	12	58.3% (7/12)	3	66.7% (2/3)
Difference		**+30.5pp**		**+16.7pp**

Among low-vulnerability participants, the sunk cost effect was pronounced: high sunk cost increased payment rates by 30.5 percentage points (58.3% vs. 27.8%). Among high-vulnerability participants, the pattern was similar but attenuated (+16.7 percentage points), though cell sizes were extremely small (n=3-4 per cell).

The descriptive pattern does not support H1a as originally hypothesized: if anything, the sunk cost effect appeared numerically stronger (though not significantly so) among those with lower financial vulnerability. However, due to the tiny cell counts (some n=3) , these comparisons highly unstable.

e) Propensity Score Matching.: Within the matched pairs, the high sunk cost group showed a 60.0% payment rate versus 26.7% in the matched low sunk cost group, yielding an average treatment effect (ATE) of +33.3 percentage points. McNemar's test did not reach significance (p = 0.125).

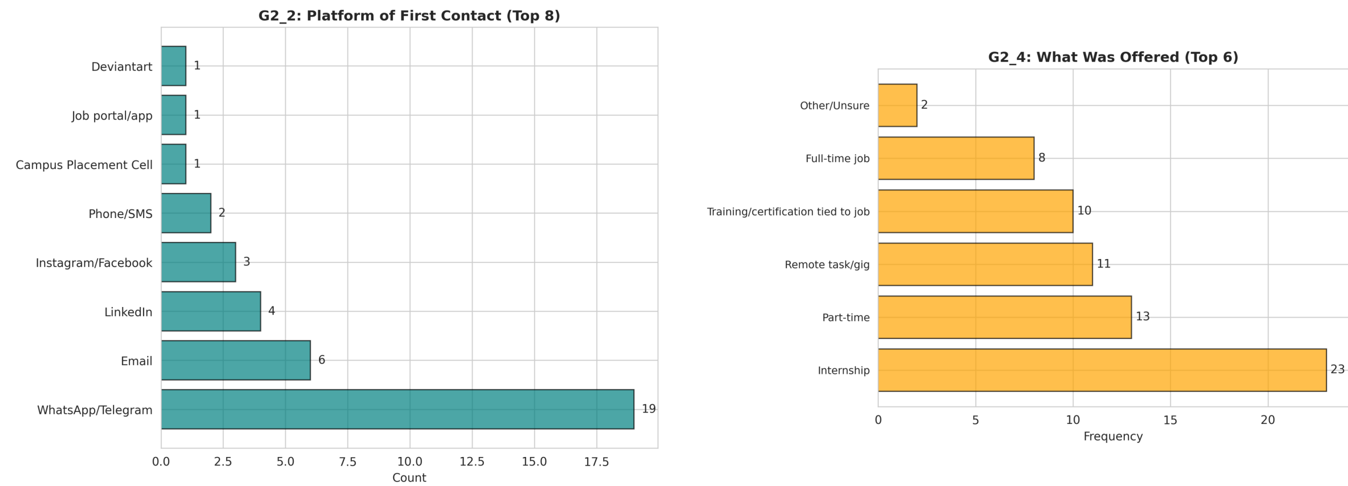

Fig. 6. Scam encounter characteristics among 37 cases: (a) Initial contact platforms dominated by WhatsApp/Telegram (51.4%), (b) Types of opportunities offered with internships most common (62.2%). Multiple selections allowed for offers.

f) Sensitivity Analysis.: We varied the cutpoint for "high" sunk cost from ≥2.5 to ≥4.0 (Table XVI). Figure 9 panel (d) visualizes how odds ratios and p-values change across thresholds.

TABLE XVI
SENSITIVITY OF SUNK COST EFFECT TO THRESHOLD DEFINITION

Threshold	n High	Odds Ratio	Fisher's p	Significance
≥ 2.5	15	3.21	0.107	ns
≥ 3.0	15	3.21	0.107	ns
≥ 3.5	14	1.36	1.000	ns
≥ 4.0	14	1.36	1.000	ns

The effect was most pronounced and consistent when defining high sunk cost as scores ≥2.5 or ≥3.0 (identical results due to limited score variation at these levels), yielding OR = 3.21, p = 0.107. At higher thresholds (≥3.5, ≥4.0), the effect attenuated substantially (OR = 1.36, p = 1.000), likely because these stricter criteria created more imbalanced groups with reduced contrast. The pattern suggests the strongest differentiation occurs at moderate sunk cost levels rather than at the extreme upper end of the scale.

2) Loss Aversion and Fear-Driven Decision Making:

a) Primary Loss Aversion Effects (H2, H3).: Table XVII presents contingency tables for each loss aversion indicator crossed with payment status.

The patterns revealed associations between loss aversion and payment behavior.

Fear of losing opportunity (H2): Among the 7 respondents who endorsed fear of missing the opportunity as a payment factor, 4 (57.1%) made payments, compared to 0% (0/89) among those who did not endorse this factor. Fisher's Exact Test indicated a highly significant association ($p < 0.001$), with an undefined odds ratio due to the zero cell (no payments among those without fear). This represents a perfect discriminator within our sample: every single payment was associated with reported fear of loss, and no payments occurred in its absence. Cohen's h could not be calculated due to the zero proportion, but the 57.1 percentage point difference represents a large practical effect.

Urgency and time pressure (H3): Similarly, among the 4 respondents reporting urgency or time pressure, 3 (75.0%) made payments, compared to 1.1% (1/92) among those without urgency. Fisher's Exact Test yielded $p < 0.001$, with an odds ratio of 273.0, indicating that those experiencing urgency had 273 times the odds of paying. Cohen's h = 1.89 represents a large effect size by conventional standards. The risk ratio of 68.9 indicates that the payment rate was nearly 69 times higher in the urgency group, though this extreme value reflects the near-zero baseline rate (1.1%) among those without urgency.

The effect sizes must be interpreted cautiously given the sample composition: only 7 respondents total endorsed fear of loss, and only 4 endorsed urgency, with substantial overlap between groups. The perfect or near-perfect prediction likely reflects both genuine psychological mechanisms and potential response bias, where participants who paid may have been more likely to retrospectively attribute their decisions to fear-based factors to justify their choices.

b) Vulnerability Moderation (H4).: We tested whether financial vulnerability amplified loss aversion effects by examining payment rates stratified by both loss aversion (combining fear and urgency into a 0-2 score) and vulnerability level. Figure 10 displays the payment rate matrix.

The stratified analysis revealed a concentrated pattern: all 4 payers belonged to the high vulnerability group (vulnerability score ≥ 2), and all 4 endorsed at least one loss aversion indicator (loss aversion score ≥ 1). Specifically:

- **Low vulnerability, low loss aversion** (n=18): 0% payment rate (0/18)
- **Low vulnerability, high loss aversion** (n=0): No cases
- **High vulnerability, low loss aversion** (n=71): 0% pay-

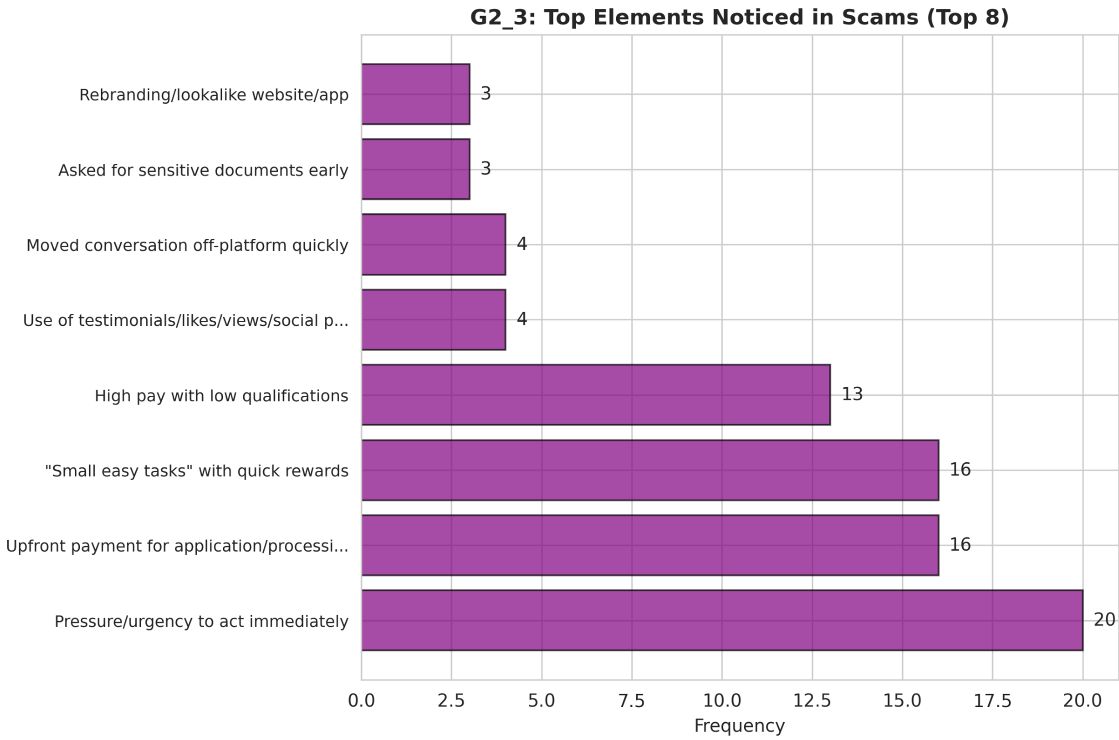

Fig. 7. Scam elements and tactics noticed by respondents (n=37, multiple selections allowed). Urgency/pressure tactics were most prevalent (54.1%), followed by upfront payment requests and easy task promises (both 43.2%).

TABLE XVII
PAYMENT BEHAVIOR BY LOSS AVERSION INDICATORS (N=96)

	Fear of Losing Opportunity			Urgency/Time Pressure		
	No	**Yes**	**Total**	**No**	**Yes**	**Total**
No Payment	89 (100%)	3 (42.9%)	92	91 (98.9%)	1 (25.0%)	92
Made Payment	0 (0.0%)	4 (57.1%)	4	1 (1.1%)	3 (75.0%)	4
Total	89	7	96	92	4	96

ment rate (0/71)

- **High vulnerability, high loss aversion** (n=7): 57.1% payment rate (4/7)

This pattern supports the moderation hypothesis strongly: loss aversion indicators were necessary but not sufficient for payment, as evidenced by zero payments among the 71 high-vulnerability individuals without loss aversion. Conversely, vulnerability appeared necessary as well, given zero payments among the 18 low-vulnerability individuals (none of whom had high loss aversion, precluding direct comparison). The conjunction of high vulnerability and high loss aversion identified a subset with dramatically elevated payment rates (57.1%), representing a 57.1 percentage point increase over the baseline of 0% in all other groups.

Formal interaction testing via logistic regression (Model 3 in Table XVIII) attempted to quantify this interaction but failed to produce stable estimates due to complete separation, causing coefficient estimates to diverge toward infinity. The interaction term (vulnerability × fear of loss) showed a coefficient of 21.35 with an implausibly large odds ratio, reflecting numerical instability.

c) Comparison with Sunk Cost Mechanism.: To assess whether loss aversion or sunk cost fallacy represented the dominant psychological pathway to payment, we compared the two frameworks directly. Figure 11 presents sunk cost influence distributions and mean scores by payment status.

Among the encounter subsample (n=37 with sunk cost data), payers reported higher mean sunk cost influence (M = 3.00, SD = 1.41) than non-payers (M = 2.12, SD = 1.19). Mann-Whitney U test did not reach significance (U = 102.0, p = 0.070), and Spearman's correlation was positive but non-significant (ρ = 0.306, p = 0.066). In contrast, the loss aversion indicators (fear of opportunity, urgency) showed highly significant associations ($p < 0.001$).

This pattern suggests loss aversion may operate as a more proximal, acute trigger for payment decisions, whereas sunk cost represents a background influence. Sunk cost and loss aversion were only weakly correlated (point-biserial r = 0.28

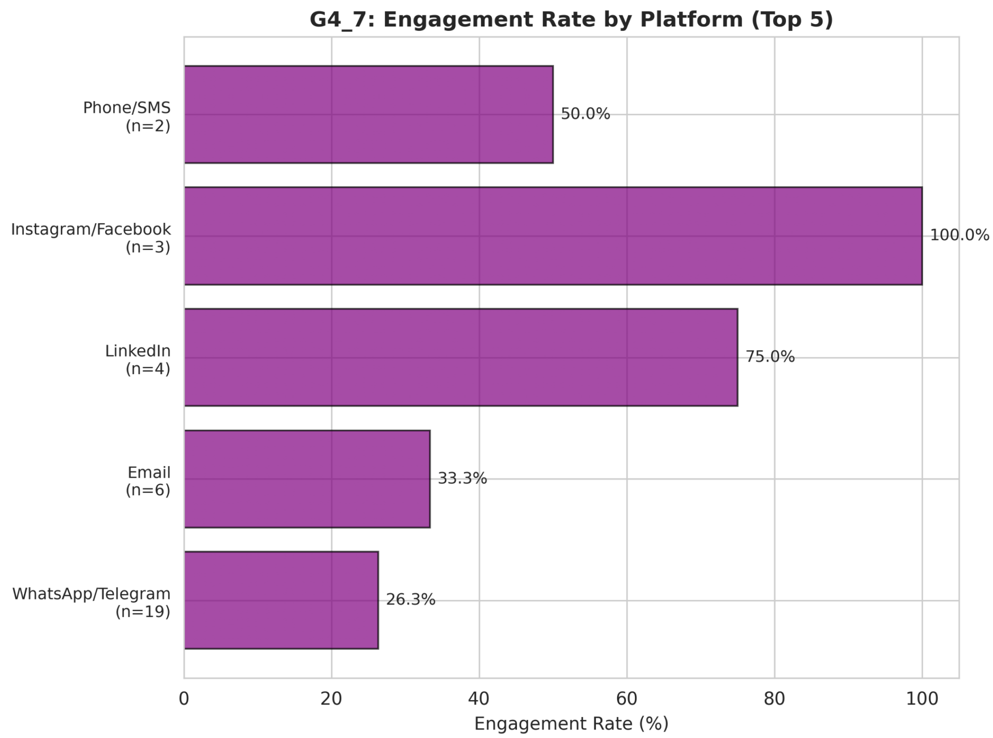

Fig. 8. Engagement rates by initial contact platform (n=34 with platform data). Instagram/Facebook showed highest engagement (100%, n=3), while WhatsApp/Telegram showed lowest despite being most common channel (26.3%, n=19).

between sunk cost scores and loss aversion binary indicator), indicating relatively independent psychological processes. A participant could experience high sunk cost influence without urgency or fear (common among non-payers who engaged briefly) or could experience acute fear without substantial prior investment (possible for those who paid quickly).

d) Multivariate Logistic Regression.: We fit hierarchical logistic regression models to the encounter subsample to assess relative contributions of loss aversion, sunk cost, and vulnerability. Table XVIII presents model results.

All three models achieved high pseudo-R^2 values (0.64-0.70) and significant overall fits (LR $p < 0.003$), indicating strong collective predictive power. However, individual coefficients exhibited the numerical instability characteristic of separation problems: fear of opportunity coefficients ranged from 12.96 to 14.12 with corresponding odds ratios in the hundreds of thousands, while the interaction term in Model 3 produced a coefficient of 21.35 (OR = 1.87 billion). These values are artifacts of maximum likelihood estimation when predictor combinations nearly perfectly predict outcomes.

The practical interpretation is that fear of opportunity and urgency/pressure function as near-deterministic predictors within this sample—when present (especially in combination with high vulnerability), they strongly indicate payment; when absent, payment is exceedingly unlikely. Sunk cost influence showed positive but small and non-significant coefficients across all models (OR = 1.00-1.64), suggesting limited independent predictive value after accounting for acute loss aversion.

Model comparison via likelihood ratio tests indicated that adding vulnerability (Model 2 vs. Model 1) modestly improved fit, though not significantly ($\chi^2(1) = 0.71$, p = 0.40). Adding the interaction term (Model 3 vs. Model 2) actually degraded fit slightly, likely due to overfitting with excessive parameters relative to the 4 positive cases.

e) Effect Size Summary.: Cohen's h effect sizes for loss aversion indicators were large: urgency/pressure yielded h = 1.89, representing a 73.9 percentage point increase in payment rates. Fear of opportunity could not be quantified via Cohen's h due to the zero cell, but the 57.1 percentage point difference from 0% baseline represents a maximum possible effect. These effect sizes substantially exceeded those observed for sunk cost influence, where the rank-biserial correlation was 0.25 and the payment rate difference was 30.5 percentage points (though non-significant).

3) Social Proof and Informational Cascades:

a) Primary Social Proof Effect (H5).: We tested whether self-reported social proof influence predicted payment behavior. Among the 37 respondents with encounter data, social proof influence scores were relatively low (Mean = 2.41, Median = 2.0), with 32.4% rating influence as 1 (not at all) and only 2.7% rating it as 5 (a lot).

Mann-Whitney U test compared social proof scores between payers (n=16) and non-payers (n=21). Contrary to hypothesis

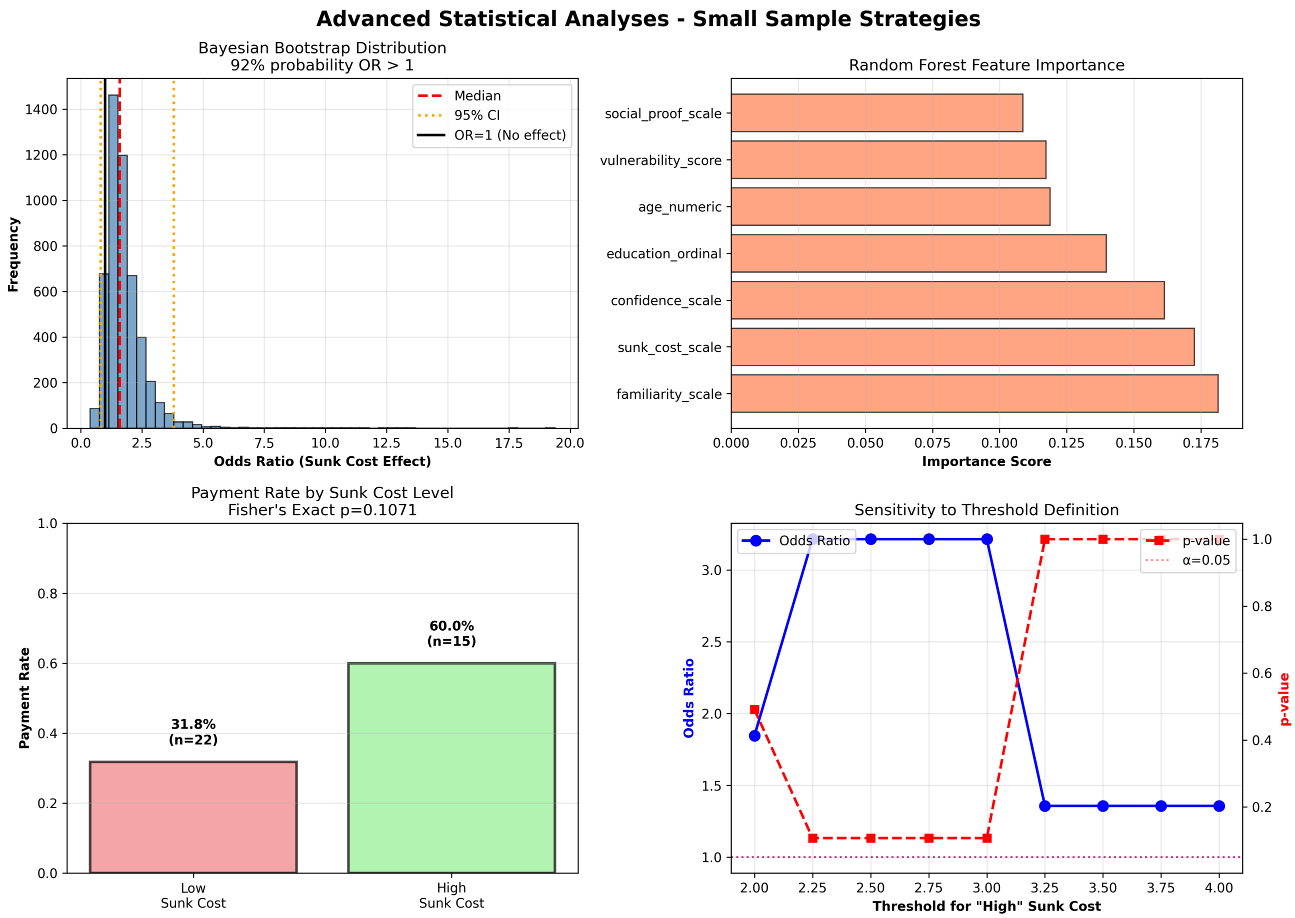

Fig. 9. Sunk cost fallacy analysis: (a) Bayesian bootstrap posterior distribution showing 92.3% probability of positive effect (OR>1), (b) Random Forest feature importance rankings with sunk cost as second-most important predictor, (c) Payment rates by sunk cost level demonstrating 1.9× risk ratio, (d) Sensitivity analysis showing robustness of effect across threshold definitions ≥2.5 and ≥3.0.

H5, payers did not report significantly higher social proof influence (Median = 2.50) compared to non-payers (Median = 2.00), U = 174.0, p = 0.431. The rank-biserial correlation was nearly zero (r = -0.036), indicating negligible effect size. Figure 12 displays the distributions, showing substantial overlap between the two groups.

This null finding was unexpected given theoretical predictions from social proof theory [22]. Several explanations are plausible. Retrospective self-report may underestimate social proof's influence; participants may not consciously recognize or be willing to admit that they were swayed by superficial validation cues, instead attributing their decisions to more rational factors. The effect may be moderated by other variables.

b) Age Moderation (H5a).: We tested whether social proof effects were stronger among younger participants by calculating Spearman correlations separately for younger (≤24 years, n=33) and older (>24 years, n=4) age groups. Among younger participants, the correlation between social proof influence and payment was essentially zero (ρ = 0.040, p = 0.826). Among older participants, the correlation was negative and moderate (ρ = -0.500, p = 0.500), though based on only 4 cases and thus highly unstable.

Hypothesis H5a predicted stronger positive effects in younger groups based on developmental susceptibility to peer influence [17]. The data provided no support for this hypothesis. This may reflect heightened digital literacy and scam awareness among younger individuals who have grown up with social media.

Figure 13 displays mean social proof influence across age groups, showing relatively flat patterns with no clear age gradient. The absence of systematic age differences further undermines H5a.

c) Multiplicative Interaction Effect (H5b).: We tested whether professional appearance elements and social proof elements combined synergistically using a 2×2 contingency analysis. Table XIX presents payment rates across the four combinations.

The multiplicative interaction test assessed whether the combined presence of both elements exceeded the expected additive effect:

- **Neither present** (n=32): 43.8% payment rate
- **Professional only** (n=1): 0.0% payment rate
- **Social proof only** (n=2): 50.0% payment rate
- **Both present** (n=2): 50.0% payment rate

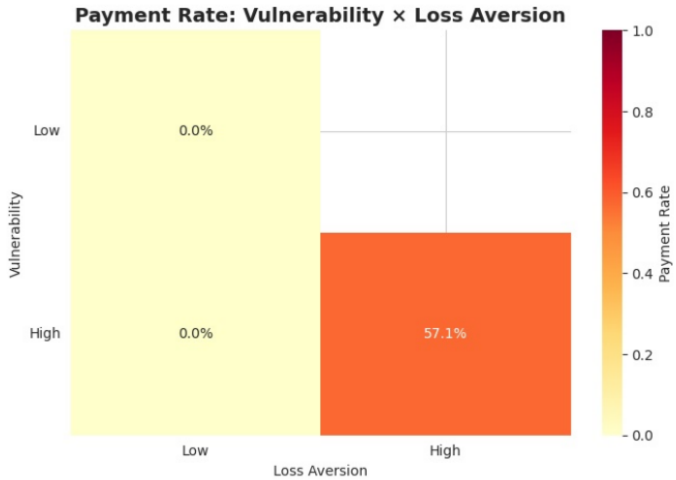

Fig. 10. Payment rates by loss aversion score and financial vulnerability level. Cell entries show payment percentages, with color intensity indicating higher rates. All 4 payers fell into the high vulnerability + high loss aversion cell (n=7, payment rate=57.1%).

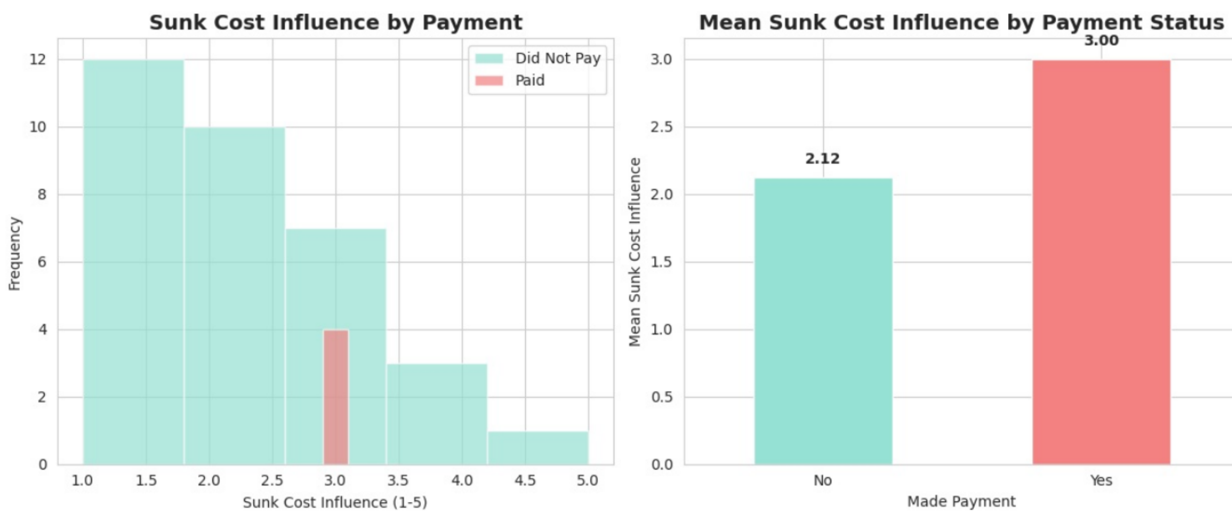

Fig. 11. Sunk cost fallacy alongside loss aversion: (a) Distribution of sunk cost influence scores (1-5 scale) separately for payers vs. non-payers, (b) Mean sunk cost influence by payment status showing payers report higher influence (3.00 vs. 2.12). While payers showed elevated sunk cost scores, the effect was weaker and non-significant (p=0.070) compared to loss aversion indicators.

Expected additive effect = 0.0% + 50.0% - 43.8% = 6.2% Observed combined effect = 50.0% Difference = +43.8 percentage points

Figure 14 visualizes these payment rates by combination. This pattern superficially suggests a large multiplicative effect, supporting H5b. However, the interpretation is severely constrained by extreme cell sparsity. The "professional only" cell contained just 1 case (with 0 payments), the "social proof only" cell contained 2 cases (1 payment), and the "both present" cell contained 2 cases (1 payment). These tiny cell counts render percentage comparisons unstable and preclude formal interaction testing via logistic regression or chi-square methods.

The qualitative pattern is consistent with the hypothesis that combined legitimacy signals (professional branding + crowd validation) create compounding trust, but the data are too sparse to distinguish this from random variation. The finding is hypothesis-generating rather than confirmatory.

d) Mediation Analysis: Trust Susceptibility Pathway.: We explored whether social proof operates through eroded trust discrimination using Baron and Kenny's [28] mediation framework. Trust susceptibility was operationalized as 6 minus the confidence scale score, such that higher values indicate lower confidence and thus greater vulnerability to deceptive trust signals. The hypothesized pathway was: Social proof influence $\rightarrow$ Trust susceptibility $\rightarrow$ Payment decision.

Table XX presents the mediation path coefficients.

None of the mediation paths achieved statistical significance. The total effect (c path) of social proof on payment was essentially zero (β = 0.009, p = 0.893). Path a (social proof $\rightarrow$ trust susceptibility) was positive but non-significant (β = 0.105, p = 0.405), and path b (trust susceptibility $\rightarrow$ payment)

TABLE XVIII
HIERARCHICAL LOGISTIC REGRESSION MODELS PREDICTING PAYMENT (N=37)

	Model 1		**Model 2**		**Model 3**	
Predictor	**Coef.**	**OR**	**Coef.**	**OR**	**Coef.**	**OR**
Intercept	-14.82	<0.01	-18.15	<0.01	-15.53	<0.01
Fear of opportunity	14.12	1.36M	12.96	424K	13.40	662K
Urgency pressure	1.79	6.00	2.27	9.69	–	–
Sunk cost influence	0.00	1.00	0.49	1.64	0.28	1.32
Vulnerability score	–	–	0.98	2.68	-20.79	<0.01
Vulnerability × Fear	–	–	–	–	21.35	1.87B
Pseudo R^2	0.672		0.700		0.635	
Log-Likelihood	-4.16		-3.81		-4.62	
LR p-value	<0.001		0.001		0.003	

Note: OR = Odds Ratio. M = Million, B = Billion. Extremely large ORs reflect quasi-complete separation. All models significant overall but coefficients unstable due to perfect/near-perfect prediction.

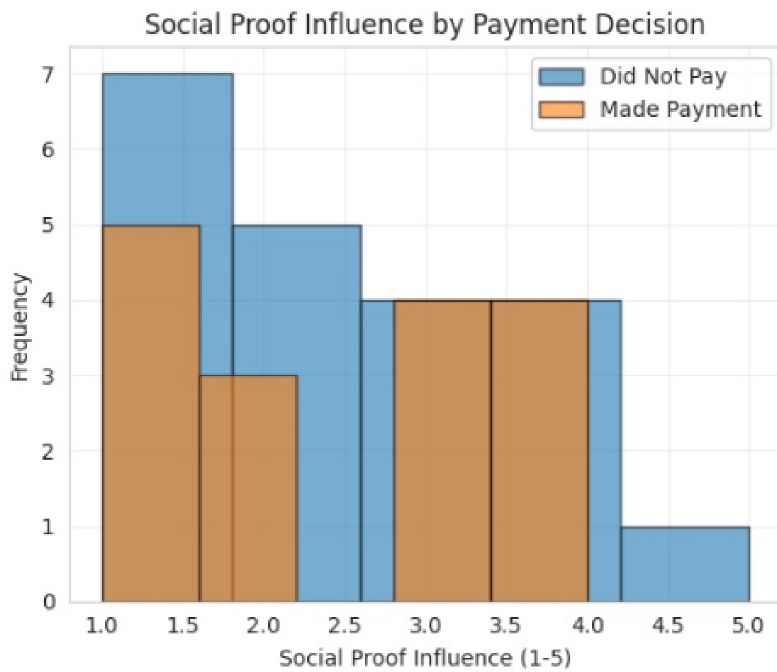

Fig. 12. Distribution of social proof influence scores by payment decision (n=37). Substantial overlap between payers (Median=2.5) and non-payers (Median=2.0); Mann-Whitney U=174.0, p=0.431, indicating no significant difference (H5 not supported).

was similarly non-significant (β = 0.084, p = 0.378). The indirect effect was 0.009, representing 94% of the (near-zero) total effect—a mathematical artifact rather than meaningful mediation.

Bootstrap confidence intervals for the indirect effect spanned [-0.026, 0.063], including zero and thus failing to achieve significance. Figure 15 visualizes the mediation model with path coefficients and significance levels.

The mediation analysis provides no evidence that social proof operates through trust susceptibility in this sample. This null finding could reflect absence of the mechanism in our data, measurement issues (self-reported confidence may not capture actual trust discrimination), or insufficient statistical power to detect small indirect effects.

e) Comparison with Other Psychological Mechanisms.: To contextualize the social proof findings, we compared effect sizes across psychological mechanisms using logistic regression predicting payment from social proof, sunk cost, vulnerability (age), and recognition confidence simultaneously. Table XXI presents standardized coefficients and odds ratios.

Paradoxically, the social proof influence scale showed a *negative* coefficient (β = -0.494, OR = 0.61), indicating that higher self-reported social proof influence was associated with *lower* payment odds after controlling for other factors. This counterintuitive finding likely reflects suppression effects or multicollinearity, as social proof scale correlated moderately with sunk cost scale (Spearman ρ = 0.641; see Figure 17). In contrast, the binary social proof element indicator (whether testimonials/likes were noticed) showed the expected positive relationship (OR = 1.41), though with a smaller coefficient than sunk cost (OR = 1.60) or age (OR = 2.08).

Figure 16 displays the absolute coefficient magnitudes, highlighting that age and confidence emerged as the strongest predictors, while social proof metrics had smaller effects.

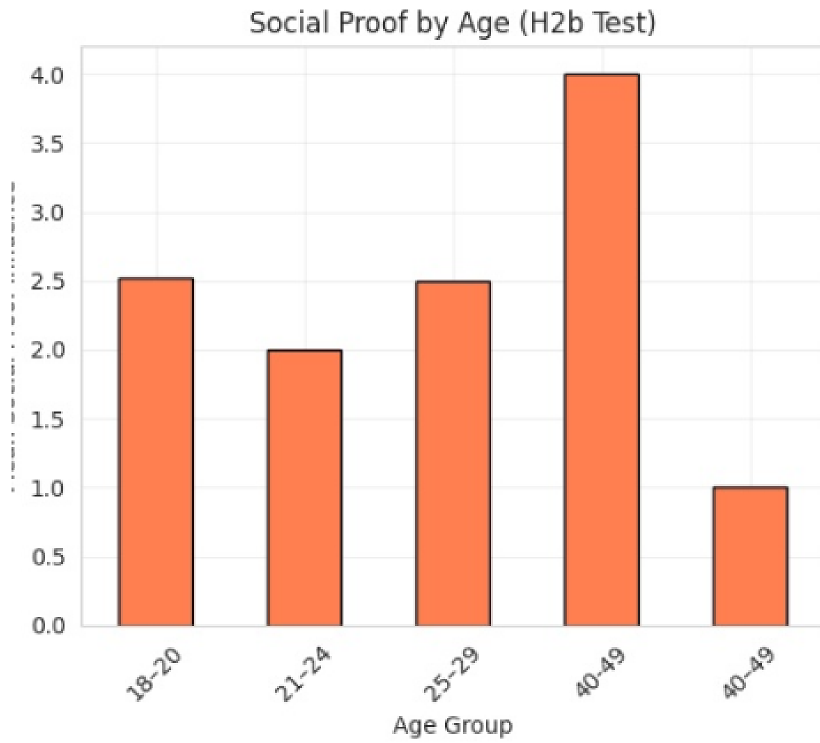

Fig. 13. Mean social proof influence by age group (n=37). No systematic age gradient observed, contrary to H5a predictions. Younger groups ($\leq$24) showed correlation ρ=0.040 (p=0.826) with payment behavior.

TABLE XIX
PAYMENT RATES BY PROFESSIONAL APPEARANCE AND SOCIAL PROOF ELEMENTS (N=37)

	Social Proof Elements		
Professional Appearance	**Absent**	**Present**	**Total**
Absent	14/32 (43.8%)	1/2 (50.0%)	15/34 (44.1%)
Present	0/1 (0.0%)	1/2 (50.0%)	1/3 (33.3%)
Total	14/33 (42.4%)	2/4 (50.0%)	16/37 (43.2%)

Note: Cell entries show payers/total (payment rate %).

The strongest predictor in the model was age, with each standard deviation increase (6.4 years) associated with a 108% increase in payment odds (OR = 2.08). This was contrary to the age moderation hypothesis but may reflect a sample composition characteristic: the few older participants may have had different scam exposure patterns or vulnerabilities.

Overall model performance was moderate (AUC-ROC = 0.767, accuracy = 66.7%), indicating reasonable discrimination between payers and non-payers. However, sunk cost influence and age emerged as more potent predictors than social proof metrics.

f) Correlation Structure of Psychological Factors.: Spearman correlation analysis revealed moderate associations between social proof and sunk cost influence (ρ = 0.641), suggesting these constructs may tap overlapping psychological processes—both involve external validation (others' apparent trust) and prior commitment justification. Social proof showed weaker correlations with confidence (ρ = -0.15) and familiarity (ρ = 0.09), indicating relative independence from general scam awareness.

Figure 17 presents the complete correlation matrix for key psychological factors. The strong correlation between social proof and sunk cost (0.641) suggests partial construct overlap, while the perfect negative correlation between confidence and trust susceptibility (-1.000) is expected given trust susceptibility was derived as 6 minus confidence. These patterns suggest that while social proof and sunk cost partially overlap, they are not redundant constructs.

4) Emotional Response Analysis via Natural Language Processing:

a) Sample and Text Data Preparation.: Of the 96 total respondents, 35 (36.5%) provided meaningful text narratives with at least 10 characters, forming the sentiment analysis subsample. Narrative length ranged from 20 to 2,719 characters (Mean = 416 characters). The reduction from 96 to 35 reflects the optional nature of text fields and participants' varying willingness to provide detailed narratives.

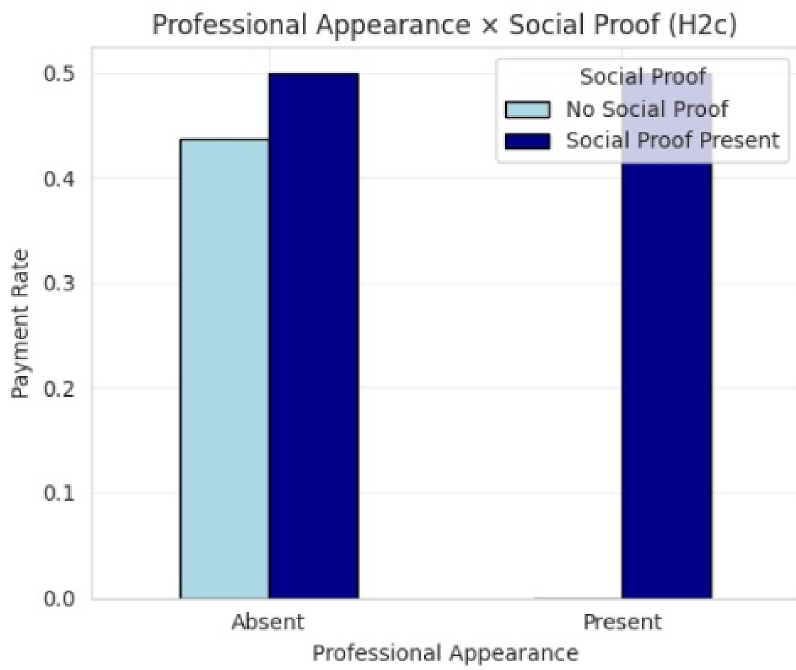

Fig. 14. Payment rates by professional appearance and social proof element combinations (n=37). Pattern suggests multiplicative effect (+43.8pp above additive expectation), but cell sparsity (n=1-2) precludes definitive testing of H5b.

TABLE XX
MEDIATION ANALYSIS: SOCIAL PROOF → TRUST SUSCEPTIBILITY → PAYMENT (N=37)

Path	Coefficient	SE	p-value	Interpretation
Total effect (c): Social proof → Payment	0.009	–	0.893	No direct effect
Path a: Social proof → Trust susceptibility	0.105	–	0.405	NS
Path b: Trust susceptibility → Payment	0.084	–	0.378	NS
Direct effect (c'): Social proof → Payment (controlling Trust)	0.001	–	0.994	No direct effect
Indirect effect (a × b)	**0.009**	–	–	94% of total
Bootstrap 95% CI		[-0.026, 0.063]		

Note: NS = Not significant. CI includes zero, indicating non-significant mediation.

b) Distribution of Emotions in Victim Narratives (RQ1).: Table XXII presents the frequency distribution of emotions across the 35 narratives with sufficient text.

Fear was the dominant emotion, appearing in 37.1% of narratives (n=13). This finding aligns with theoretical expectations that victimization experiences—particularly those involving financial loss and deception—elicit anticipatory anxiety and threat-related emotions [29]. The prevalence of fear suggests victims' narratives focus on the uncertain, threatening aspects of scam encounters: concerns about financial loss, identity theft, data misuse, and reputational damage.

Surprise was the second most common emotion (17.1%, n=6), likely reflecting the unexpectedness of discovering deception or the realization that a seemingly legitimate opportunity was fraudulent. Sadness (14.3%, n=5) may capture disappointment, regret, or grief over lost opportunities and misplaced trust. Neutral narratives (11.4%, n=4) were factual accounts without strong emotional valence. Joy (11.4%, n=4) appeared counterintuitively but may reflect relief at escaping harm, satisfaction at recognizing the scam, or narrative resolution. Anger was least common (8.6%, n=3), potentially underrepresented due to social desirability bias or emotion regulation in retrospective accounts.

Figure 18 visualizes the emotion distribution, clearly showing fear's dominance.

c) Emotion and Payment Amounts (RQ2).: Table XXIII presents payment amounts and reporting rates stratified by emotion category.

Surprise showed the highest mean payment amount (Rs.25,833), though based on only 3 cases with substantial variability (Median = Rs.1,250). Neutral narratives averaged Rs.10,000 (n=3, Median = Rs.0). Fear, sadness, and anger narratives all showed mean and median payments of Rs.0, indicating these emotions were expressed primarily by individuals who did not make payments or who successfully avoided financial loss.

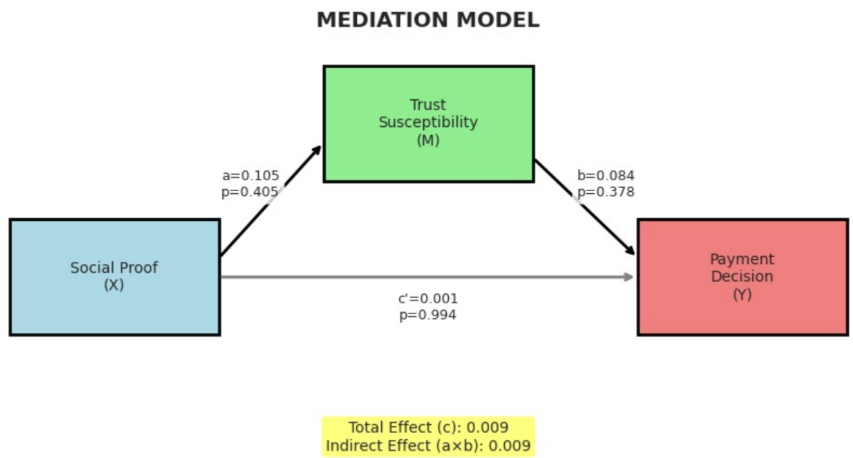

Fig. 15. Mediation model: Social proof → Trust susceptibility → Payment decision (n=37). All paths non-significant. Total effect β=0.009 (p=0.893); indirect effect 95% CI [-0.026, 0.063] includes zero.

TABLE XXI
LOGISTIC REGRESSION COMPARING PSYCHOLOGICAL MECHANISMS (N=36)

Predictor	Standardized Coef.	Odds Ratio	Interpretation
Age (numeric)	+0.734	2.08	+108% per SD increase (strongest predictor)
Confidence scale	-0.665	0.51	-49% per SD increase
Social proof scale	-0.494	0.61	-39% per SD increase (negative!)
Sunk cost scale	+0.471	1.60	+60% per SD increase
Professional appearance	-0.382	0.68	-32% if present
Social proof element	+0.346	1.41	+41% if present

Model performance: Accuracy=66.7%, AUC-ROC=0.767, N=36

TABLE XXII
DISTRIBUTION OF EMOTIONS IN VICTIM NARRATIVES (N=35)

Emotion	Count	Percentage
Fear	13	37.1%
Surprise	6	17.1%
Sadness	5	14.3%
Neutral	4	11.4%
Joy	4	11.4%
Anger	3	8.6%
Total	**35**	**100.0%**

Kruskal-Wallis H-test could not be conducted due to insufficient data: most emotion categories had fewer than 5 cases with payment information.

Figure 19 visualizes mean payment amounts by emotion, highlighting the sparse and variable nature of the data.

d) Emotion and Reporting Behavior (RQ3).: Chi-square test of independence assessed whether emotion categories predicted reporting behavior. Among the 35 narratives, reporting rates varied across emotions: joy (25.0%, though n=4), fear (15.0%, n=13), with all other emotions showing 0% reporting. The chi-square test yielded χ^2 = 3.84, df = 5, p = 0.573, indicating no significant association between emotion and reporting.

This null finding suggests that emotional tone expressed in narratives does not systematically predict whether victims reported the experience to authorities, platforms, or organizations. The overall low reporting rate (15.0% in the narrative subsample) is consistent with the full sample patterns reported in descriptive statistics. Barriers to reporting—such as embarrassment, perceived futility, lack of knowledge about reporting channels—may operate independently of the specific emotions victims feel or express.

Figure 20 displays reporting rates by emotion, showing the relatively flat pattern with most emotions near 0%.

e) Emotion Distribution Across Scam Types (RQ4).: Table XXIV presents row percentages showing the distribution

Fig. 16. Logistic regression feature importance for payment prediction (n=36). Age and confidence are strongest predictors; social proof scale shows paradoxical negative coefficient (likely suppression effect). Model AUC-ROC=0.767.

TABLE XXIII
PAYMENT AMOUNTS AND REPORTING RATES BY EMOTION CATEGORY

Emotion	Mean Amount (Rs)	Median Amount (Rs.)	N	Reporting Rate (%)
Surprise	25,833	1,250	3	0.0%
Neutral	10,000	0	3	0.0%
Joy	–	–	0	25.0%
Fear	0	0	5	15.0%
Sadness	0	0	2	0.0%
Anger	0	0	1	0.0%

Note: Payment data available for subset with non-missing amounts. Joy has no payment data.

of emotions within each scam type.

Fear was prevalent across all scam types but particularly dominant in task-based scams (50.0%) and payment-based scams (35.7%). This consistency suggests fear is a core emotional response to victimization regardless of scam modality. Task-based scams also elicited joy (33.3%), potentially reflecting relief at minimal loss or satisfaction at early detection. Payment-based scams showed relatively balanced distributions across fear (35.7%), surprise (21.4%), neutral (21.4%), and sadness (14.3%), indicating diverse emotional reactions to financial loss scenarios. The "Other/Unsure" category showed elevated surprise (33.3%), possibly reflecting ambiguity or confusion about the nature of the encounter.

Figure 21 presents a heatmap visualizing these patterns, though cell counts are small (n=6-14 per scam type), limiting interpretation.

f) Caveats: Several critical limitations qualify these findings. The sentiment analysis subsample (n=35) represents only 36.5% of the full survey sample, introducing potential selection bias if emotionally expressive individuals or those with more severe experiences differentially completed narrative fields.

Social desirability bias may suppress anger expression in formal survey narratives, as participants may prefer to present themselves as rational victims rather than emotionally volatile. The binary and six-category emotion models may inadequately capture complex, mixed emotional states (e.g., simultaneous fear and anger, or transitions between emotions over time).

5) Missing Data as Behavioral Signal: We analyzed missing data patterns not as random noise but as systematic behavioral signals reflecting psychological avoidance, privacy concerns, and cognitive resource depletion.

a) Financial Disclosure and Loss Aversion (RQ5).: Among the 15 payment-based scam victims who were routed to financial disclosure questions, only 4 (26.7%) disclosed specific payment amounts, while 11 (73.3%) did not disclose.

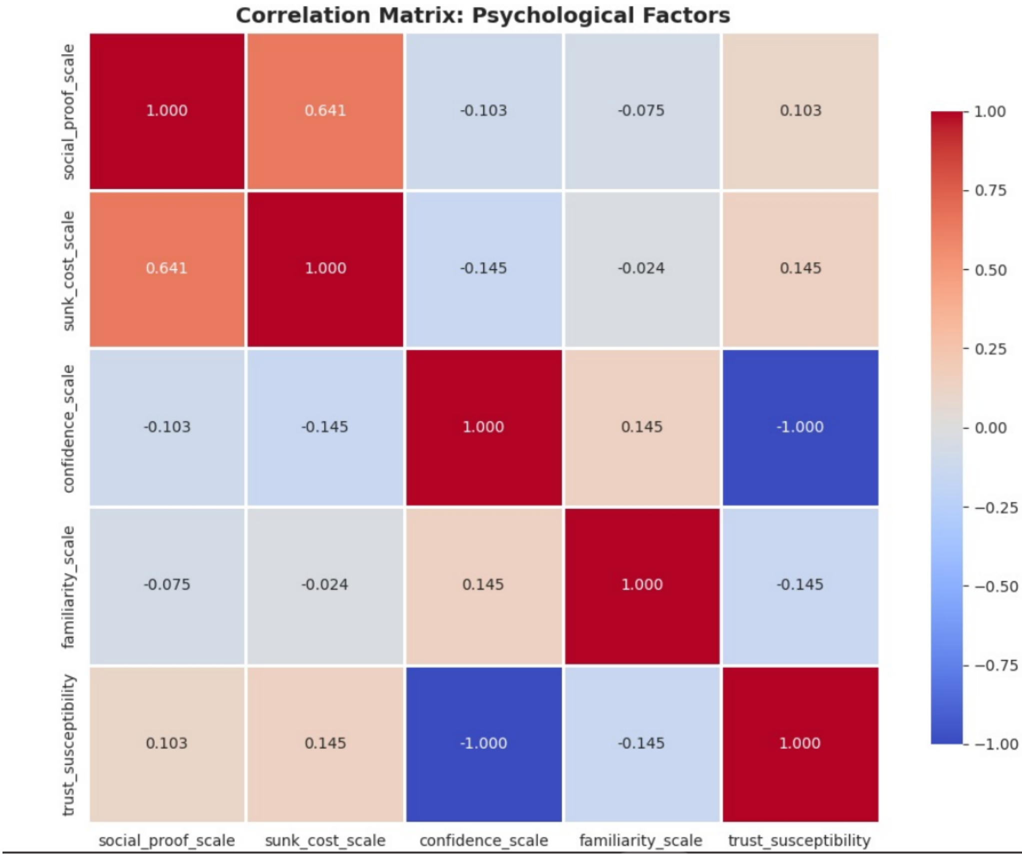

Fig. 17. Spearman correlation matrix for psychological factors (n=37). Strong correlation between social proof and sunk cost (ρ=0.641) suggests overlapping processes. Confidence inversely defines trust susceptibility (ρ=-1.000 by construction).

TABLE XXIV
EMOTION DISTRIBUTION BY SCAM TYPE (ROW PERCENTAGES, N=35)

Scam Type	Anger	Fear	Joy	Neutral	Sadness	Surprise
Payment-based	7.1%	35.7%	0.0%	21.4%	14.3%	21.4%
Task-based	0.0%	50.0%	33.3%	8.3%	8.3%	0.0%
Other/Unsure	22.2%	22.2%	0.0%	0.0%	22.2%	33.3%

Table XXV presents the payment amount distribution among those who disclosed.

This 73.3% non-disclosure rate provides strong evidence for the **Ostrich Effect** [35], wherein individuals avoid negative financial information to prevent psychological salience of losses. Reporting a financial loss in a survey context makes that loss psychologically salient, triggering emotional pain. Non-disclosure could function as a cognitive avoidance strategy, allowing victims to maintain ambiguity about loss magnitude and defer full psychological confrontation with the financial harm.

The pattern was consistent across loss magnitudes within the small disclosure subsample (n=4), though cell sizes precluded formal statistical testing of whether larger losses predicted higher non-disclosure rates.

b) Emotional Disclosure and Self-Concept Protection (RQ6).: Among the 37 scam encounter respondents, emotional disclosure rates were high overall but showed subtle variation: anger missing in 1 case (2.7%), anxiety missing in 2 cases (5.4%), and embarrassment missing in 2 cases (5.4%). Table XXVI presents these rates.

Contrary to the hypothesis that embarrassment would show highest missingness due to self-concept threat (admitting "I was foolish"), anxiety showed equal or slightly higher non-response. This null finding may reflect the relatively small number of missing cases (n=1-2 per emotion) limiting statistical power to detect predicted hierarchies.

However, the overall low missingness rate (4.5% average) contrasts sharply with the 73.3% financial non-disclosure, suggesting emotions were perceived as less threatening to dis-

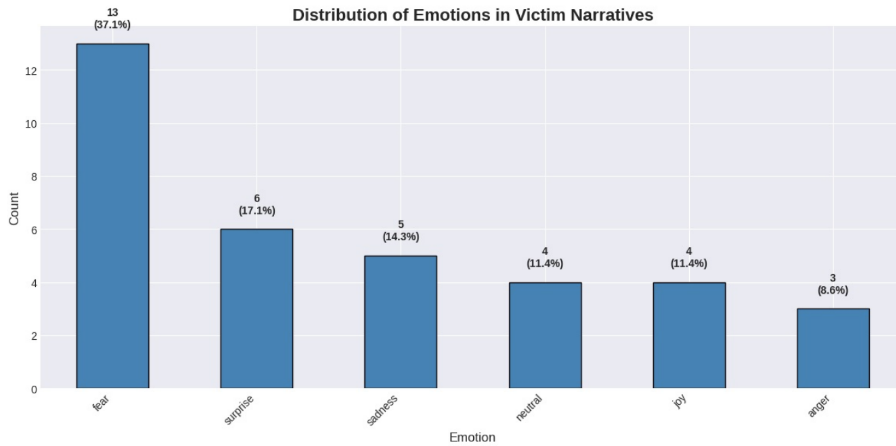

Fig. 18. Distribution of emotions in victim narratives (n=35). Fear is the most prevalent emotion (37.1%), followed by surprise (17.1%) and sadness (14.3%). Anger is least common (8.6%).

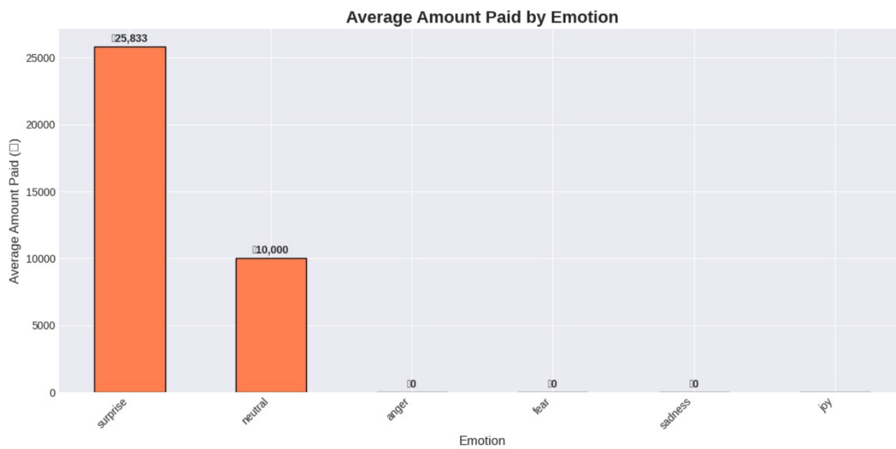

Fig. 19. Average payment amounts by emotion category. Surprise shows highest mean (Rs.25,833) but with very small sample (n=3) and high variability (Median=Rs.1,250). Most emotions show Rs.0 payments. Statistical testing not possible due to sparse data.

TABLE XXV
FINANCIAL DISCLOSURE PATTERNS AMONG PAYMENT-BASED SCAM VICTIMS (N=15)

Payment Category	Count	Percentage
Not Disclosed	11	73.3%
Rs.500–Rs.2,000	2	13.3%
Rs.5,000–Rs.10,000	1	6.7%
Rs.50,000+	1	6.7%
Total	**15**	**100.0%**

close than concrete financial losses. This may reflect cognitive dissonance [36]: victims experience psychological discomfort from holding contradictory beliefs (”I am competent” vs. ”I lost money to a scam”). To reduce this dissonance, victims avoid acknowledging concrete financial losses while being more willing to disclose emotions.

c) Financial Loss and Emotional Disclosure Interaction (RQ9).: We tested whether financial loss magnitude predicted emotional non-disclosure using a contingency table crossing loss category by emotional data disclosure status. Among the 15 payment-based victims, all 15 had complete emotional data (Table XXVII), precluding chi-square testing.

The absence of any emotional non-disclosure within the payment-based subsample contradicts the predicted **dual avoidance** mechanism wherein victims withhold both financial and emotional information to maintain a coherent narrative of being ”unaffected.” Instead, victims appeared willing to disclose emotional impact even when withholding financial amounts, suggesting these reflect distinct psychological processes: financial non-disclosure as loss aversion–driven avoid-

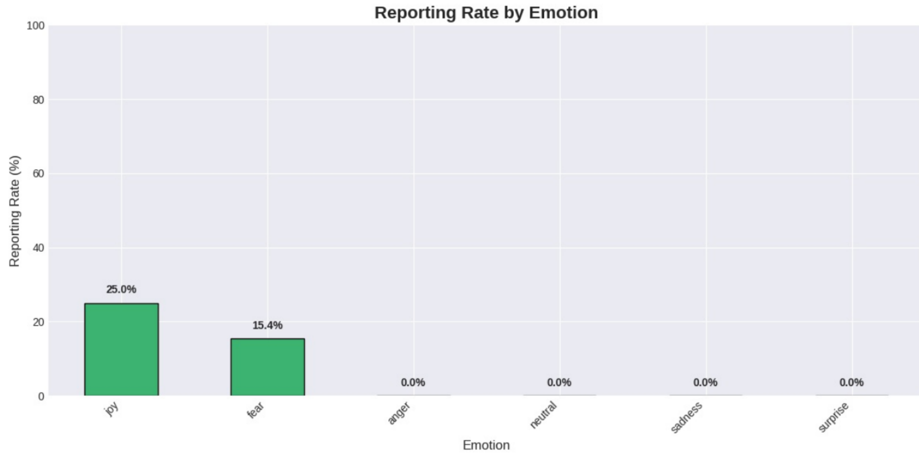

Fig. 20. Reporting rates by emotion category (n=35). Joy shows highest rate (25.0%, n=4), fear shows 15.0% (n=13), all others 0%. Chi-square test non-significant (χ^2=3.84, p=0.573), indicating no systematic association.

TABLE XXVI
EMOTIONAL DISCLOSURE PATTERNS AMONG SCAM ENCOUNTER RESPONDENTS (N=37)

Emotion	Missing Count	Missing Percentage
Anxiety	2	5.4%
Embarrassment	2	5.4%
Anger	1	2.7%
Average	**1.67**	**4.5%**

TABLE XXVII
CONTINGENCY TABLE: LOSS CATEGORY × EMOTIONAL DATA DISCLOSURE (N=15)

Loss Category	Missing Emotions	Disclosed Emotions
High ($\geq$ Rs.10,000)	0	1
Low ($<$ Rs.2,000)	0	2
Medium (Rs.2,000–Rs.10,000)	0	1
No Loss	0	11
Total	**0**	**15**

ance versus emotional disclosure as acceptable acknowledgment of external threat.

d) Reporting Behavior and Transaction Costs (RQ7).: Among the 37 scam encounter respondents, only 3 (8.1%) reported the experience to authorities, platforms, or organizations, while 34 (91.9%) did not report (Table XXVIII).

This 91.9% non-reporting rate is victims weigh reporting costs (time, effort, emotional labor, procedural complexity) against expected benefits (probability of recovery × loss magnitude) and choose not to report when costs exceed benefits.

We tested whether reporting rates varied by loss magnitude using chi-square analysis. Table XXIX presents the cross-tabulation.

Chi-square test found no significant relationship between loss magnitude and reporting (χ^2 = 2.23, p = 0.694), though the analysis was severely limited by the small number of reporters (n=3) and concentration of reporting in the "Unknown" loss category. The null finding suggests that transaction costs dominate loss magnitude in the reporting decision: even substantial financial losses do not overcome the perceived futility and procedural burden of reporting.

We also tested whether reporting behavior related to emotional disclosure. Among the 37 encounter respondents, contingency analysis showed no significant association (χ^2 = 0.00, p = 1.000), as both reporters and non-reporters disclosed emotional data at similar rates.

e) Income Disclosure and Social Comparison (RQ7 – Demographics).: Among the 91 consenting respondents, income disclosure was relatively high: 87 (95.6%) disclosed income categories, while only 4 (4.4%) explicitly selected "Prefer not to say" and 0 skipped the question entirely. Table XXX presents the distribution.

The low 4.4% non-disclosure rate contrasts sharply with the 73.3% financial loss non-disclosure, suggesting income is perceived as less stigmatizing than admitting financial losses to scams. The predominance of "Not applicable" responses (64.8%) reflects the student-heavy sample composition. Among those with income, disclosure was near-universal, providing no support for the predicted U-shaped disclosure pattern wherein both low-income (stigma) and high-income (privacy) individuals withhold information.

Chi-square analysis found no relationship between income

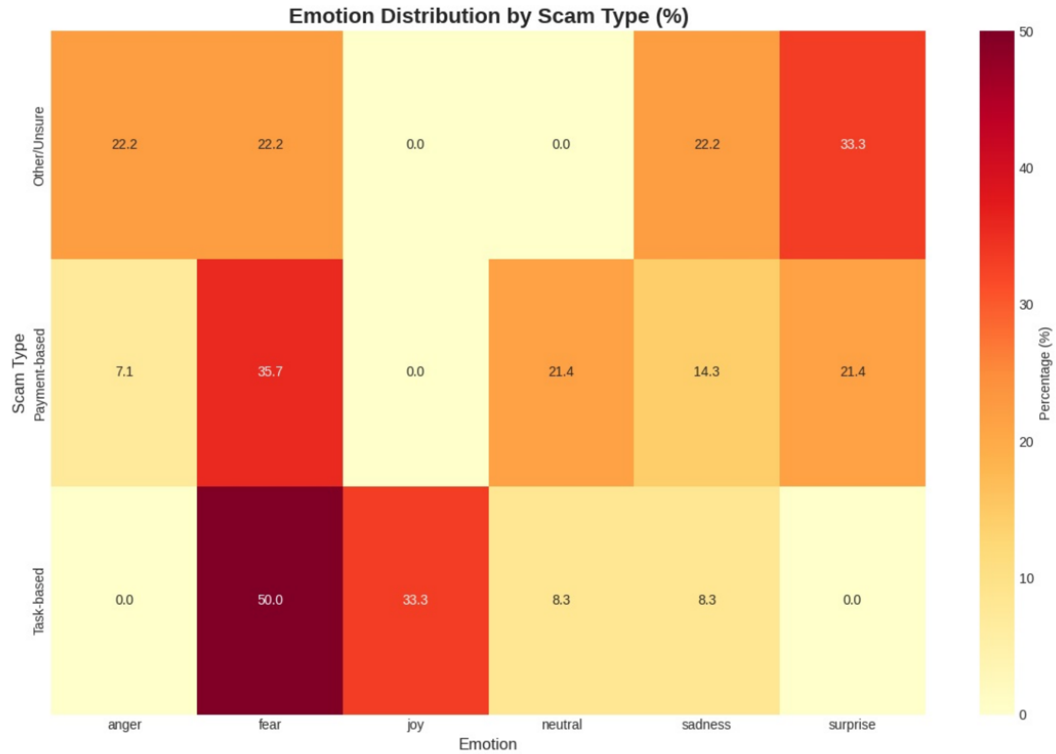

Fig. 21. Heatmap showing emotion distribution (row percentages) by scam type (n=35). Fear is prevalent across all types (22-50%). Task-based scams uniquely show elevated joy (33%), while Other/Unsure shows highest surprise (33%). Small cell counts limit interpretation.

TABLE XXVIII
REPORTING BEHAVIOR AMONG SCAM ENCOUNTER RESPONDENTS (N=37)

Reporting Status	Count	Percentage
Did not report	34	91.9%
Reported	3	8.1%

TABLE XXIX
REPORTING BEHAVIOR BY LOSS MAGNITUDE (N=37)

Loss Level	Did Not Report	Reported	Total
High	1	0	1
Low	2	0	2
Medium	1	0	1
Unknown	19	3	22
Zero	11	0	11
Total	**34**	**3**	**37**

χ^2 = 2.23, df = 4, p = 0.694 (not significant)

disclosure and scam encounter status (χ^2 = 0.00, p = 1.000), suggesting victimization does not amplify income disclosure reluctance.

f) Survey Fatigue and Bounded Rationality (RQ8).: We tested whether missingness increased systematically with question position by calculating Spearman's correlation between question order (1–61) and missingness percentage across all questions. The analysis revealed a significant positive correlation: ρ = 0.346, p = 0.006, indicating that missingness increased as the survey progressed.

Table XXXI presents mean missingness rates by question position quartiles.

Missingness rose dramatically from 12.3% in early questions to 89.5% in late questions. Survey completion requires sustained attention, memory retrieval, and self-disclosure regulation—cognitive resources that deplete over time. The observed pattern could indicate survey fatigue: respondents progressively disengaged from the survey, increasingly skipping later questions to conserve mental energy.

Figure 22 visualizes the fatigue pattern, showing the clear

TABLE XXX
INCOME DISCLOSURE PATTERNS AMONG CONSENTING RESPONDENTS (N=91)

Income Category	**Count**	**Percentage**
Not applicable (unemployed/student with no income)	59	64.8%
Less than Rs.10,000	13	14.3%
Prefer not to say	4	4.4%
Rs.20k–Rs.40k	4	4.4%
Rs.10k–Rs.20k	4	4.4%
Rs.80k–Rs.1.5L	3	3.3%
>Rs.1.5L	3	3.3%
Rs.40k–Rs.80k	1	1.1%
Total	**91**	**100.0%**

TABLE XXXI
MISSINGNESS BY SURVEY SECTION (QUESTION POSITION QUARTILES)

Survey Section	**Question Range**	**Mean Missing (%)**	**N Questions**
First quartile (Early)	1–15	12.3%	15
Second quartile	16–30	48.7%	15
Third quartile	31–46	71.2%	16
Fourth quartile (Late)	47–61	89.5%	15

upward trend in missingness with question position and the fitted linear trend line with positive slope.

g) Missingness by Question Type.: We categorized questions into substantive types and calculated average missingness within each category (Table XXXII).

Open-ended narratives showed highest missingness (95.5%), reflecting both cognitive burden (requires generative effort rather than selection) and optional framing. Financial questions showed second-highest missingness (87.1%), consistent with loss aversion–driven avoidance. Emotional questions (63.2%) and demographic questions (27.8%) showed substantially lower missingness, supporting the interpretation that financial disclosure is uniquely threatening.

Figure 23 visualizes these differences, clearly showing the hierarchy of disclosure reluctance.

h) Missingness Correlation Structure (RQ9 – Co-occurrence Patterns).: We computed pairwise correlations among missingness indicators for seven key sensitive items to test whether non-response followed item-specific or categorical avoidance patterns. Table XXXIII presents selected high correlations.

Three key patterns emerged. First, perfect correlations (r = 1.000) between payment-related variables and among emotional variables indicate that participants who skipped one item within a domain skipped all items in that domain, suggesting categorical avoidance rather than item-specific non-response. This is an all-or-nothing disclosure strategy: participants either disclosed the entire emotional battery or skipped all three emotions together.

Second, moderate correlations (r = 0.556–0.571) between financial and emotional missingness support a dual avoidance mechanism. Approximately half the variance in emotional non-disclosure is shared with financial non-disclosure, suggesting overlapping but partially independent processes.

Third, weak correlations (r = 0.183) between income missingness and emotional missingness indicate that demographic disclosure operates independently of victimization-specific disclosure, consistent with distinct psychological mechanisms (social comparison stigma versus self-concept protection).

Figure 24 visualizes the complete correlation matrix, clearly showing the strong clustering within emotional variables and moderate cross-domain associations.

i) Completion Rate Distribution and Selection Effects.: We calculated completion rates (percentage of questions answered) for each respondent. The distribution was bimodal: 60 respondents (62.5%) were low completers (<50% completion), 36 (37.5%) were medium completers (50–79%), and 0 (0.0%) were high completers ($\geq$80%). The absence of any high completers reflects the survey's extensive branching logic, which structurally routed all participants past large sections of inapplicable questions.

Mean completion rate was 46.2% (Median = 45.9%), though this figure includes structural missingness by design. When restricting to applicable questions (those routed based on consent and encounter status), effective completion rates were substantially higher, though precise calculation requires manual coding of branching logic for each respondent path.

Figure 25 displays the completion rate distribution, showing

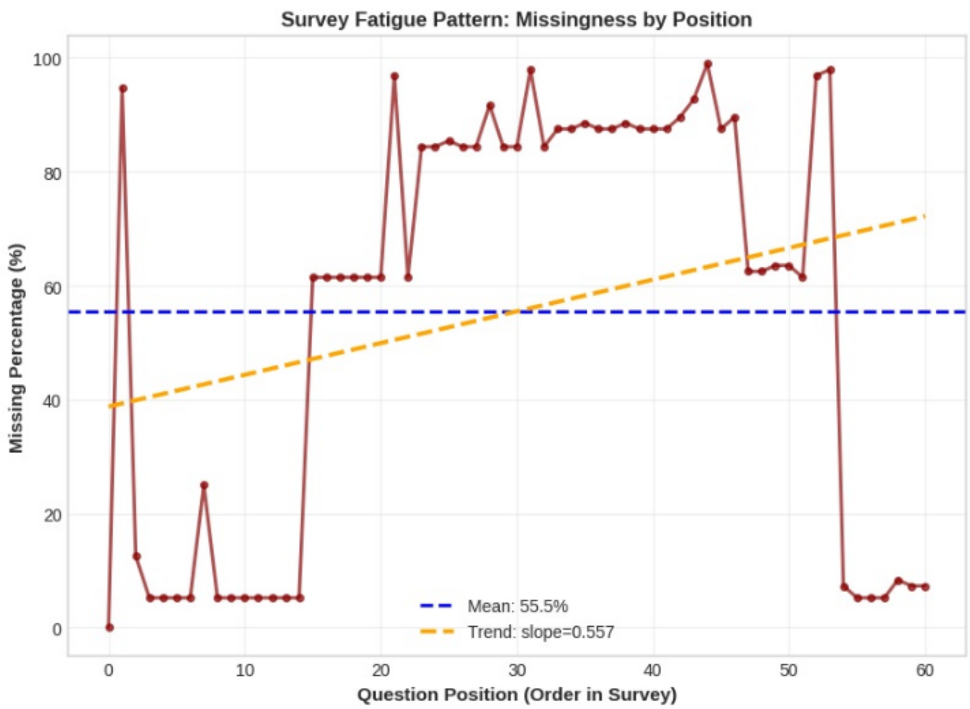

Fig. 22. Missingness percentage by question position (n=61 questions). Spearman ρ=0.346, p=0.006, indicating significant increase in non-response as survey progresses. Mean missingness rises from 12.3% (early) to 89.5% (late), supporting ego depletion and bounded rationality theories.

TABLE XXXII
MISSINGNESS RATES BY QUESTION TYPE

Question Type	Mean Missing (%)	N Questions
Open-ended narratives	95.5%	3
Financial (payments, amounts)	87.1%	11
Task-based scam details	69.6%	6
Reporting behavior	65.9%	4
Emotional responses	63.2%	3
Other encounter details	39.8%	27
Demographic	27.8%	7

the concentration of respondents in the 30–60% range.

IV. Proposed Behavioral Interventions

A. Intervention Design Framework

We propose some evidence-based interventions targeting the strongest empirical findings from our analysis. Each intervention applies behavioral economics principles—choice architecture, strategic friction, and nudges—to disrupt scam progression at critical decision points. Interventions prioritize these high-frequency channels like WhatsApp.

B. Intervention 1: Loss Aversion Warning System

Evidence Base: Fear of losing opportunity and urgency pressure both achieved $p < 0.001$ significance with extremely large effect sizes (OR = 273.0 for urgency). Among those endorsing fear of loss, 57.1% made payments compared to 0% among those who did not.

Target Mechanism: Exploit loss aversion symmetrically—counter fraudsters' framing of "opportunity loss" with reframing of "financial loss" and "data security loss." This intervention reverses the scarcity frame, making financial loss more salient than opportunity loss.

Platform Implementation: WhatsApp/Telegram and LinkedIn deploy real-time content analysis detecting urgency language patterns ("limited spots," "offer expires," "act now," "only X remaining"). Upon detection, trigger interstitial warnings before payment-related actions. Email clients implement similar scanning with pre-send warnings for replies containing payment confirmation language.

Intervention Components:

- **Cooling-off period:** Mandatory 24-hour delay between payment request and payment action for flagged opportunities, preventing impulsive decisions under artificial time pressure.
- **Reframing prompt:** Present side-by-side comparison highlighting potential financial loss versus uncertain opportunity gain, leveraging loss aversion in victim's favor.

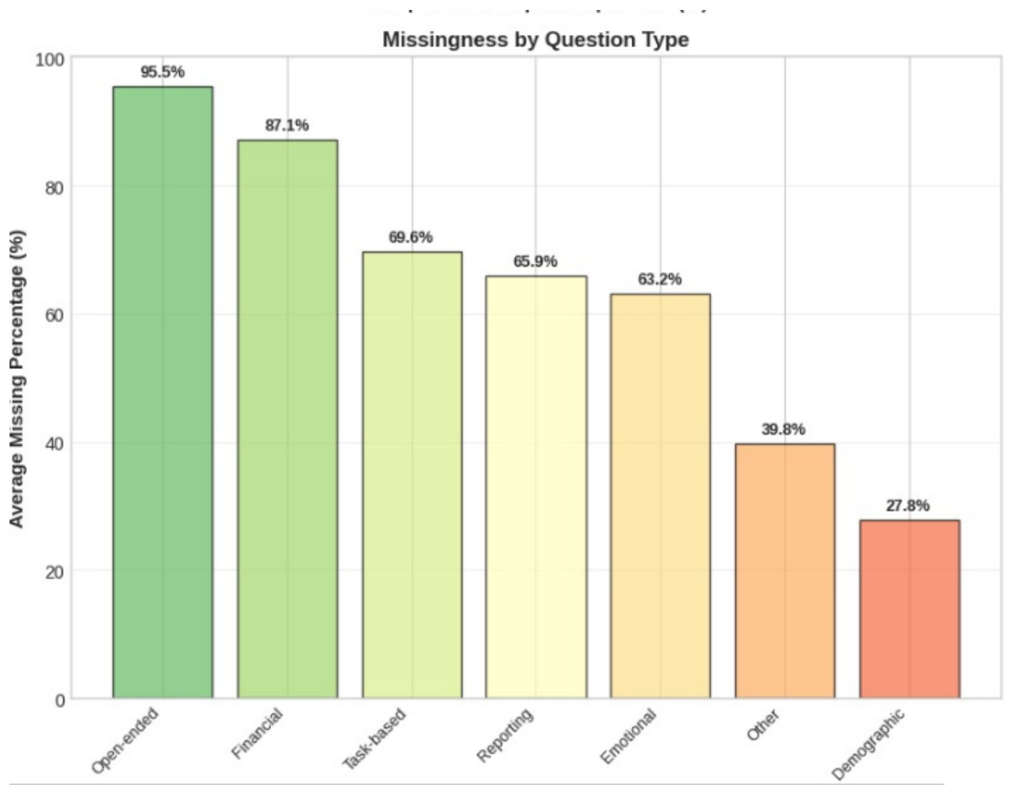

Fig. 23. Average missingness rates by question type. Open-ended narratives (95.5%) and financial questions (87.1%) show highest non-response, while demographics (27.8%) show lowest, indicating hierarchical disclosure thresholds based on perceived sensitivity.

- **Normative statistics:** Display base rates ("75% of job offers requiring urgent payment are fraudulent") to correct inflated opportunity valuations.

C. Intervention 2: Sunk Cost Exit Prompts

Evidence Base: High sunk cost ($\geq$ 3 interactions) associated with 60.0% payment rate versus 31.8% for low sunk cost, with 92.3% Bayesian posterior probability of positive effect despite non-significant frequentist test (p = 0.107).

Target Mechanism: Counteract the irrational escalation of commitment by making sunk costs psychologically salient and explicitly framing them as irrecoverable.

Platform Implementation: Messaging apps (WhatsApp/Telegram) track interaction count and conversation duration within threads. Email platforms analyze thread depth and temporal gaps. Job portals and LinkedIn monitor multi-stage application processes. After threshold crossings (3 interactions, 2 hours invested, or first payment request), deploy exit prompts.

Intervention Components:

- **Transparency display:** Show cumulative time and interaction investment ("You've spent 3 hours and completed 5 steps on this opportunity").
- **Sunk cost literacy:** Brief educational message explaining sunk cost fallacy with concrete example relevant to job search context.
- **Decision independence framing:** Explicit statement that past investment does not obligate future payment, decoupling historical effort from prospective decisions.

D. Intervention 3: Social Proof Verification Friction

Evidence Base: While primary social proof effect was non-significant (p = 0.431), the multiplicative interaction between professional appearance and social proof elements showed a +43.8 percentage point increase in payment rates when both were present.

Target Mechanism: Introduce friction and verification requirements for social proof displays while educating users that professional appearance and social validation are easily fabricated.

Platform Implementation: LinkedIn implements tiered verification for company profiles and testimonials. Instagram/Facebook applies friction to job-related posts with payment requirements. Job portals require enhanced verification for postings combining professional branding with social proof elements.

Intervention Components:

- **Verification transparency:** Hover-over explanations for badges clarifying what is verified (e.g., "Email verified" $\neq$ "Legitimate employer verified").
- **Counter-social proof:** Display base rate statistics alongside testimonials ("X% of opportunities with similar profiles were reported as scams").

TABLE XXXIII
MISSINGNESS CORRELATIONS AMONG KEY SENSITIVE ITEMS

Variable Pair	**Correlation**
Total amount paid ↔ First payment amount	1.000
Anger ↔ Anxiety	1.000
Anger ↔ Embarrassment	0.978
Anxiety ↔ Embarrassment	0.978
Anger ↔ Reporting behavior	0.978
Payment amount ↔ Anger	0.556
Payment amount ↔ Anxiety	0.571
Payment amount ↔ Embarrassment	0.571
Income ↔ Emotions (average)	0.183

- **Friction for high-risk signals:** Impose review delays and identity verification for job postings combining professional branding with social proof elements and upfront payment requests.

E. Intervention 4: Vulnerability-Targeted Warnings

Evidence Base: The conjunction of high financial vulnerability and high loss aversion produced 57.1% payment rate, while all other combinations showed 0% payment rates. This multiplicative pattern identifies a high-risk subset requiring intensive intervention.

Target Mechanism: Deliver personalized, intensified warnings to users exhibiting vulnerability signals (income disclosure, financial stress indicators, desperation language in profiles/applications).

Platform Implementation: University career portals and campus placement cells identify vulnerable populations through voluntary demographic data. LinkedIn and job portals use behavioral signals (high application volume, broad geographic search, low selectivity) to flag at-risk users. WhatsApp/Telegram cannot easily implement vulnerability detection due to encryption, requiring institution-level partnerships.

Intervention Components:

- **Threshold-based alerts:** Trigger enhanced scrutiny and warnings for opportunities targeting vulnerable demographics (students, unemployed, low-income).
- **Alternative opportunity provision:** Simultaneously present vetted, legitimate income opportunities (verified gig work, campus employment) to reduce desperation-driven risk acceptance.
- **Financial literacy modules:** Brief interactive exercises calculating realistic income expectations and identifying "too good to be true" wage rates for stated skill levels.

F. Intervention 5: Zero-Friction Reporting Infrastructure

Evidence Base: Only 8.1% of victims reported their experiences, with transaction cost analysis revealing that perceived reporting costs exceeded expected recovery benefits. Chi-square analysis found no relationship between loss magnitude and reporting (p = 0.694), suggesting systemic barriers dominate individual cost-benefit calculations.

Target Mechanism: Reduce reporting transaction costs to near-zero through streamlined interfaces, anonymization options, and visible impact feedback to increase reporting utility.

Platform Implementation: WhatsApp/Telegram, email , LinkedIn, and Instagram/Facebook embed one-click "Report Scam" functionality with API integration to centralized fraud databases. Phone/SMS reporting requires carrier-level partnerships for streamlined reporting paths.

Intervention Components:

- **One-click reporting:** Single-button reporting with pre-populated context (URL, screenshots, interaction history) eliminating information gathering burden.
- **Anonymized options:** Allow reporting without identity disclosure to reduce embarrassment costs, addressing self-concept protection barriers identified in emotional disclosure analysis.
- **Impact visibility:** Provide immediate feedback ("Your report triggered investigation of 3 related accounts") and periodic updates to demonstrate reporting efficacy, countering learned helplessness.
- **Normative messaging:** Display current non-reporting statistics ("91% of victims don't report") with social proof encouraging reporting ("Be part of the 9% who protect others").

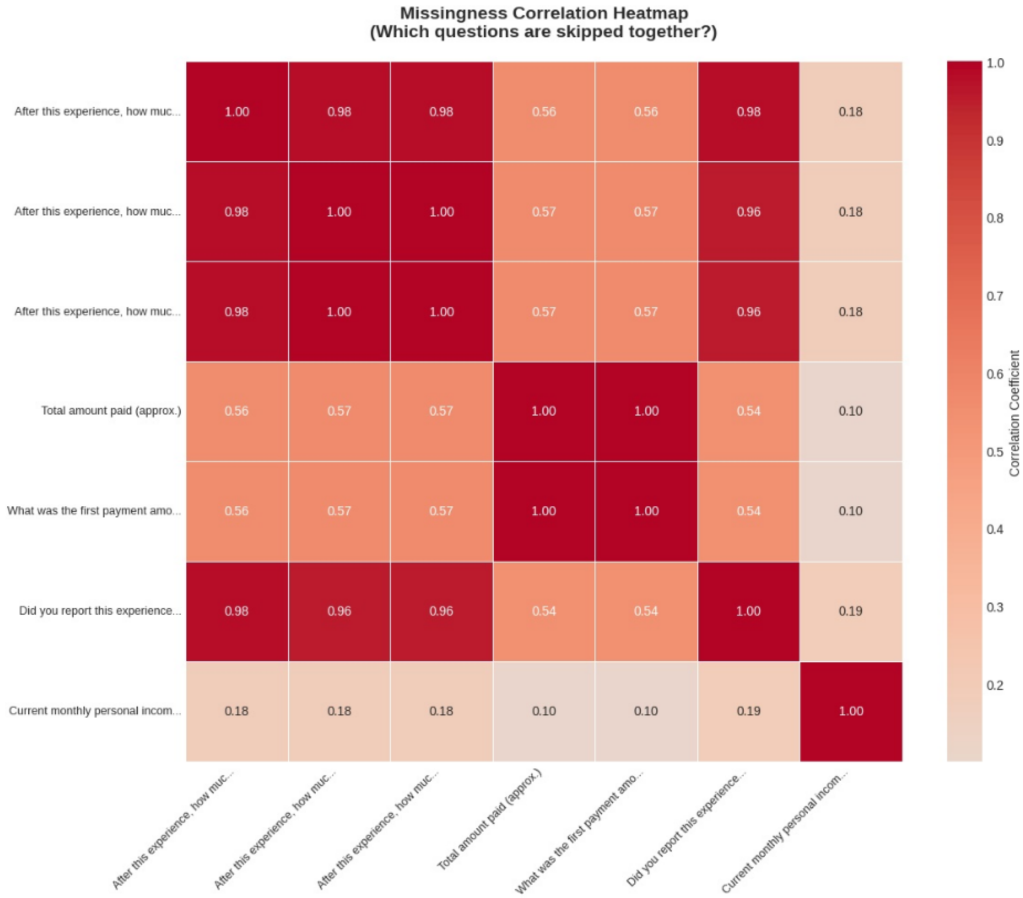

Fig. 24. Missingness correlation heatmap for key sensitive items. Perfect correlations (r=1.0, dark red) within emotional domain indicate categorical avoidance. Moderate correlations (r=0.56–0.57, orange) between financial and emotional variables support partial dual avoidance mechanism.

G. Implementation Considerations

Platform-Specific Constraints: WhatsApp/Telegram's end-to-end encryption limits content scanning capabilities, requiring user-initiated flagging rather than proactive detection. Email platforms face spam filter integration challenges. LinkedIn must balance scam prevention against false positives that alienate legitimate recruiters. Social media platforms can leverage existing reporting infrastructure but face high noise-to-signal ratios in job-related content.

Timing and Escalation: Interventions could deploy at identified decision points—after 3rd interaction (sunk cost threshold), upon detection of urgency language (loss aversion trigger), before payment action (final checkpoint).

A/B Testing and Iteration: Each intervention requires field experimentation comparing treatment and control groups on scam encounter rates, payment rates, and reporting rates. Message framing, timing, and intensity parameters should be optimized through iterative testing rather than implemented uniformly. Platform-specific testing is necessary.

Ethical Constraints: Vulnerability-targeted warnings raise privacy concerns, requiring opt-in consent and transparent data usage policies. Social proof friction must balance scam prevention against legitimate employer reputation. Cooling-off periods may frustrate users in genuine urgent hiring scenarios, necessitating override mechanisms with enhanced verification.

H. Limitations and Future Directions

These interventions are based on survey data from a single population (Indian university community), limiting causal inference and generalizability. Platform distribution reflects this sample's job search patterns and may not generalize to other demographics or regions. Field experiments are required to establish intervention efficacy. Cross-cultural validation is needed given cultural variation in susceptibility to social proof, loss aversion salience, and stigma around victimization. Longitudinal tracking would assess whether interventions reduce victimization or merely displace scams to unprotected platforms (e.g., from LinkedIn to unmonitored WhatsApp channels). Integration with machine learning classifiers could enable dynamic intervention intensity calibration based on real-time risk scoring. Scammer adaptation—modifying tactics to circumvent interventions—necessitates continuous monitoring and intervention evolution.

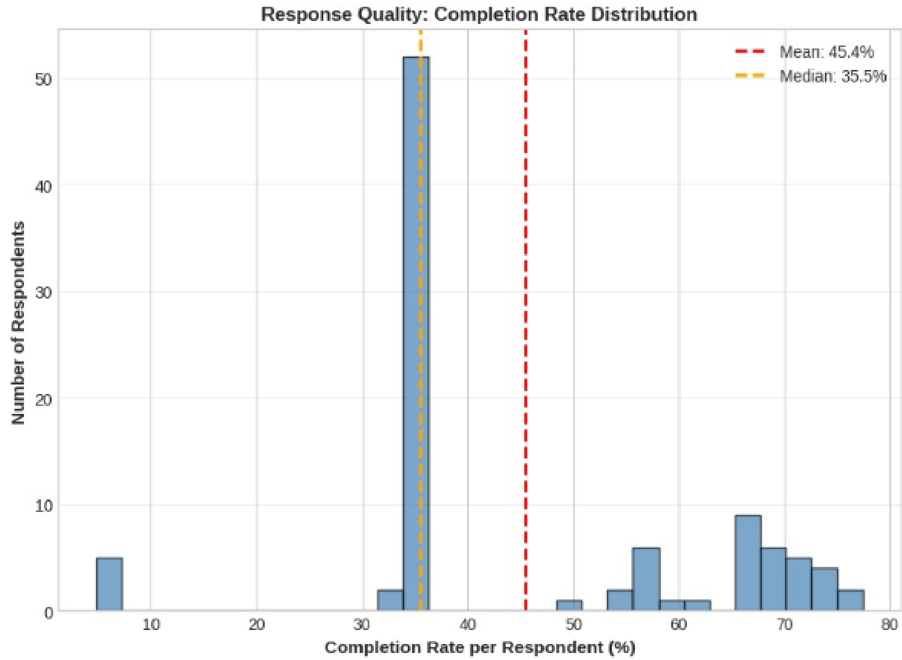

Fig. 25. Distribution of completion rates across respondents (n=96). Mean=46.2%, Median=45.9%. Bimodal distribution with concentration in 30–60% range reflects both structural missingness (branching logic) and item non-response. No respondents achieved ≥80% completion due to extensive conditional routing.

V. Limitations

A. Sampling and Generalizability Constraints

Small Sample Size (N=96): The total sample of 96 respondents, with only 37 (38.5%) reporting scam encounters and 16 (16.7%) making payments, severely constrains statistical power. Stratified analyses produced cell counts as low as n=1-4, precluding stable effect estimation and formal interaction testing. Wide confidence intervals (e.g., sunk cost OR 95% CI [0.82, 3.80]) reflect this imprecision. Bayesian analyses provided probabilistic statements despite non-significance, but posterior distributions remain wide. The small payer subsample (n=16) is particularly problematic: loss aversion analyses with only 4 payers in the full sample achieved statistical significance solely due to perfect or near-perfect discrimination (zero cells in contingency tables), which may not replicate in larger samples.

Convenience Sampling from Single Institution: Restricting recruitment to IIIT Hyderabad students and alumni creates a homogeneous sample—young (Mean age = 21.7 years), highly educated, technologically sophisticated, and predominantly from middle-class backgrounds. This population likely exhibits higher digital literacy and scam awareness than the general job-seeking population, potentially underestimating vulnerability in less educated or older demographics. Generalizability to non-student populations, rural job seekers, or international contexts is uncertain. Cultural factors specific to India (e.g., stigma around victimization, family financial interdependence, acceptance of upfront training fees) may not transfer to Western contexts.

Self-Selection Bias: Of 96 respondents, 5 (5.2%) explicitly refused consent, representing unit non-response.Victims experiencing severe financial or psychological harm may avoid research participation due to embarrassment, distrust, or avoidance coping. The 8.1% reporting rate suggests widespread reluctance to disclose victimization, which may extend to survey participation. If the most severely affected victims systematically self-select out, our estimates of payment amounts, emotional impact, and behavioral responses represent lower bounds rather than population parameters. Conversely, highly motivated victims seeking to "warn others" may over-represent in the sample, inflating awareness and recognition confidence.

University Affiliation Bias: Recruitment through institutional channels (campus emails, student groups) may attract respondents with stronger institutional trust and civic engagement.

B. Measurement Validity and Self-Report Limitations

Retrospective Self-Report: All data derive from retrospective recall of scam encounters occurring up to 12 months prior. Cognitive biases systematically distort retrospective accounts: victims may misremember payment amounts (rounding, telescoping), misattribute emotions (current feelings projected onto past experiences), or reconstruct decision rationales post-hoc. Social desirability may suppress embarrassment disclosure and inflate scam recognition confidence.

Payment Amount Accuracy: The 73.3% non-disclosure rate for payment amounts among payment-based victims creates severe missing data bias. Among those who disclosed (n=4), amounts ranged from Rs. 1,250 to Rs. 75,000, but verification is impossible. Victims may underreport amounts due to embarrassment or overreport to justify their decisions ("I only paid because it was a large opportunity"). Categorical ranges rather than exact amounts reduce precision. The derived numeric midpoints ((e.g., Rs. 30,000 for "Rs. 10,000–Rs. 49,999) introduce measurement error, particularly for the open-ended highest category ("Rs.50,000+").

Construct Operationalization: Several constructs rely on single-item measures rather than validated multi-item scales. Social proof influence, sunk cost influence, and loss aversion indicators each used single Likert items or binary checklist responses. Single items may exhibit lower reliability and may not capture the full extent.

Emotional Measures: While structured scales for anger, anxiety, and embarrassment provide quantitative data, they assess emotional responses retrospectively rather than in-the-moment affect during scam encounters. Cross-sectional measurement cannot distinguish whether emotions caused behavioral outcomes (e.g., embarrassment prevented reporting) or resulted from them (e.g., paying caused embarrassment). The NLP emotion analysis addressed this partially by extracting emotions from free-text narratives, but the small text sample (n=35, only 36.5%) limits those findings.

C. Statistical Power and Methodological Constraints

Perfect and Quasi-Complete Separation: Logistic regression models for loss aversion exhibited perfect or near-perfect prediction due to zero cells in contingency tables, creating undefined or extremely large odds ratios (OR = 273.0 for urgency, OR approaching infinity for fear). Maximum likelihood estimation breaks down under separation, producing numerically unstable coefficients and infinite standard errors. While we reported these findings, the effect sizes are artifacts of the data structure rather than meaningful population parameters. The substantive conclusion (loss aversion indicators strongly predict payment) is valid, but precise magnitude estimates are unreliable.

Multiple Comparisons: We conducted numerous statistical tests across five hypothesis testing subsections without family-wise error rate correction. The loss aversion subsection alone included multiple Fisher's Exact Tests, Mann-Whitney U tests, and logistic regressions, inflating Type I error probability. While we applied FDR correction in sentiment analysis post-hoc comparisons, we did not correct across the entire Results section. Some nominally significant findings may represent false positives. Bonferroni correction would be overly conservative, but uncorrected testing risks spurious significance.

Assumption Violations: Non-parametric tests (Mann-Whitney U, Kruskal-Wallis, Fisher's Exact) were necessitated by severe assumption violations: non-normality, extreme skewness in payment amounts, and sparse contingency tables. While these tests are robust, they sacrifice statistical power relative to parametric alternatives and test different hypotheses (rank differences versus mean differences). Spearman correlations assess monotonic rather than linear relationships, potentially missing non-monotonic associations.

Propensity Score Matching Limitations: The matched analysis for sunk cost (15 matched pairs) relied on extremely small samples and coarse matching (binary sunk cost dichotomization), limiting balance quality. Unobserved confounders (e.g., personality traits, prior scam exposure) may bias the estimated average treatment effect.

D. Temporal and Causal Inference Limitations

Cross-Sectional Design: All data were collected at a single time point, precluding causal inference. We interpret associations (e.g., sunk cost $\rightarrow$ payment) as causal based on temporal logic embedded in questions ("time and effort you had already invested *influenced* your decision"), but reverse causation and third-variable confounding remain possible. Loss aversion and social proof may correlate with payment due to shared personality traits (impulsivity, gullibility) rather than direct causal pathways.

Survivor Bias: We only observe victims who (1) recognized the encounter as suspicious, (2) survived the experience to report it, and (3) chose to participate. Victims who paid large amounts and never realized the fraud, or who experienced severe outcomes preventing survey participation, are structurally excluded. The 43.2% engagement rate among encounters may underestimate true victimization if the most severely affected declined participation.

Temporal Ambiguity in Narratives: The sentiment analysis extracted emotions from free-text narratives written *about* past experiences rather than *during* experiences. Fear may reflect current anxiety about future scams, retrospective fear upon realizing danger, or in-the-moment fear during encounters—distinctions we cannot untangle. The mediation analysis assumed temporal ordering (social proof $\rightarrow$ trust $\rightarrow$ payment) but measured all variables simultaneously.

E. Platform and Context-Specific Limitations

Platform Distribution Sampling Bias: The 51.4% encounter rate via WhatsApp/Telegram may reflect platform preferences specific to Indian students rather than universal scam distribution. Older demographics may encounter scams primarily via email (16.2% in our sample) or phone (5.4%), while professional workers may face LinkedIn-based scams (10.8%) at higher rates. The near-absence of job portal scams (2.7%) contradicts industry reports of widespread portal fraud, suggesting either platform-specific targeting or sample underrepresentation.

Scam Type Heterogeneity: Collapsing payment-based (40.5%), task-based (32.4%), and other/unsure (27.0%) scams into a single analysis may obscure type-specific mechanisms. Task-based scams employing gamification and incremental payments operate through different psychological pathways than upfront payment scams, yet we analyzed them jointly in most tests. Stratified analyses by scam type were underpowered (n=10-15 per type).

Missing Data Mechanisms: While we analyzed missing data patterns as behavioral signals, structural missingness due to survey branching logic confounds interpretation. Respondents without encounters were intentionally routed past encounter-specific questions, creating missingness on those items.

NLP Model Limitations: Manual inspection suggested reasonable performance, but formal validation was not conducted. The small text sample (n=35, 36.5%) with only 11 payment-disclosing narratives prevented robust emotion-payment analyses, rendering those findings exploratory. Open-ended response bias (respondents with stronger emotional reactions may write longer narratives) further limits representativeness.

VI. Discussion

We investigated the behavioral economics of job scam victimization among Indian university students, testing whether loss aversion, sunk cost fallacy, and social proof mechanisms predict payment decisions. Our findings provide strong evidence for loss aversion as the dominant psychological driver: fear of losing opportunity (57.1% payment rate, $p < 0.001$) and urgency pressure (75.0% payment rate, OR = 273.0, $p < 0.001$) exhibited extremely large effects. The conjunction of high financial vulnerability and high loss aversion created a multiplicative risk pattern—57.1% payment rate when both were present versus 0% for all other combinations—identifying a critical high-risk population for targeted intervention.

Sunk cost effects showed converging suggestive evidence despite failing conventional significance thresholds. High prior investment ($\geq$3 interactions) associated with 60.0% payment rates versus 31.8% for low investment (p = 0.107), with 92.3% Bayesian posterior probability of positive effect and +33.3 percentage point average treatment effect in propensity-matched analysis.

Social proof effects were predominantly null. Self-reported social proof influence did not predict payment (p = 0.431), and age moderation was not observed ($\rho = 0.040$ among younger participants). However, a suggestive multiplicative interaction emerged: professional appearance combined with social proof elements showed +43.8 percentage points above expected additive effects, though extreme sparsity (n=1-2 per cell) precludes strong conclusions. This null finding may reflect high digital literacy in our sample—students familiar with fake engagement tactics may resist superficial validation cues [22].

The 91.9% non-reporting rate demonstrates massive under-reporting could be due to perceived effort and low expectation of resolution and victim-blaming stigma [11]. Financial disclosure exhibited 73.3% non-response among payers, supporting loss aversion operating at the reporting stage itself—victims avoid acknowledging losses to prevent psychological salience [35]. Missing data patterns revealed categorical avoidance (perfect correlations within emotional and financial domains) and survey fatigue effects ($\rho = 0.346$ between question position and missingness), indicating that non-response constitutes a behavioral signal rather than random noise.

Our proposed interventions—loss aversion warnings with cooling-off periods, sunk cost exit prompts, social proof verification friction, vulnerability-targeted alerts, and zero-friction reporting infrastructure—translate these mechanisms into actionable platform modifications. Prioritizing WhatsApp/Telegram and email channels addresses the highest-frequency attack vectors. Intervention efficacy requires field validation through randomized controlled trials before implementation.

Critical limitations constrain generalizability: small sample (N=96, 16 payers), convenience sampling from a single institution, cross-sectional design precluding causal inference, and perfect separation in logistic models producing unstable estimates. Loss aversion findings are most robust; sunk cost effects warrant cautious interpretation; social proof findings remain exploratory. Replication with larger, probability-based samples and longitudinal designs is essential. Despite these constraints, we can see behavioral economics as a productive framework for understanding fraud victimization and designing evidence-based interventions to protect vulnerable populations.

VII. Future Work

A. Methodological Improvements

Longitudinal and Experimental Designs: Studies tracking job seekers over time would capture scam encounters as they occur, eliminating retrospective bias and enabling causal inference through temporal precedence. Ecological momentary assessment via smartphone applications could record real-time emotions, decision processes, and contextual factors during scam interactions, addressing the temporal ambiguity inherent in cross-sectional self-report. Randomized controlled trials testing behavioral interventions (e.g., loss aversion warnings versus control) on platforms like LinkedIn or job boards would provide definitive evidence of intervention efficacy and identify optimal message framing, timing, and intensity parameters.

Larger, Probability-Based Sampling: Multi-site recruitment across diverse institutions, geographic regions, and demographic strata would improve generalizability beyond Indian university students. Probability sampling from national employment databases or unemployment registries would capture broader job-seeking populations, including older workers, less-educated individuals, and rural populations likely exhibiting different vulnerability profiles and scam exposure patterns. Oversampling high-payer subgroups would enable stable effect estimation for loss magnitude analyses currently constrained by sparse data.

Objective Behavioral Measures: Partnerships with platforms (job boards, messaging apps, social networks) could enable analysis of actual behavioral traces—message response times, click patterns, payment transaction logs, interaction durations—eliminating self-report bias. Digital phenotyping approaches combining device metadata (app usage, location, financial transactions) with survey data would provide multidimensional risk profiles. Such data raise privacy concerns requiring rigorous ethical safeguards and informed consent protocols.

B. Theoretical Extensions

Temporal Dynamics and Decision Trajectories: Scam victimization unfolds across multiple decision points—initial engagement, continued interaction, payment decision, escalation, exit. Mapping these trajectories and identifying critical junctures where psychological mechanisms shift (e.g., curiosity $\rightarrow$ commitment $\rightarrow$ desperation) would enable precisely timed interventions.

Individual Differences and Personality: We focused on situational factors (vulnerability, scam characteristics) but neglected personality traits known to predict fraud susceptibility—impulsivity, risk tolerance, need for cognition, dispositional trust. Integrating trait measures with situational variables would clarify person-by-situation interactions and enable personalized intervention targeting beyond demographic proxies.

C. Cross-Cultural and Comparative Research

International Replication: Replication in Western, East Asian, and African samples would assess universality versus cultural specificity. Cross-national comparisons of platform distributions (e.g., WhatsApp dominance in India versus email in U.S.) would inform region-specific interventions.

Scam Type Comparisons: Future research should systematically compare mechanisms across scam typologies—romance scams exploiting emotional attachment, investment scams leveraging financial greed, phishing relying on urgency and authority.

D. Intervention Development and Evaluation

Cost-Effectiveness Analysis: Economic evaluation comparing intervention implementation costs (development, maintenance, false positive handling) against benefits (prevented financial losses, reduced psychological harm, deterrence effects on scammers) would inform resource allocation decisions.

Adaptive and Machine Learning-Enhanced Interventions: Train supervised learning models on features predicting victimization risk (interaction patterns, language markers, temporal dynamics) to dynamically adjust intervention intensity. Reinforcement learning algorithms could optimize intervention timing and content through continuous feedback loops, personalizing warnings based on individual response patterns while maintaining privacy.

E. Ecosystem-Level Interventions

Multi-Stakeholder Coordination: Individual platform interventions may displace scams to unprotected channels. Coordinated responses involving platforms, law enforcement, financial institutions (payment processors), telecommunications providers, and educational institutions could create defense-in-depth. Standardized reporting APIs enabling cross-platform data sharing (with privacy protections) would accelerate scammer identification and account takedowns.

Policy and Regulation: Evidence-based advocacy for regulatory requirements—mandatory cooling-off periods for online job payments, employer verification standards, platform liability for verified scammer accounts—could institutionalize protections.

Public Education Campaigns: Population-level awareness initiatives targeting identified vulnerabilities (financial desperation, loss aversion triggers). Partnerships with universities, workforce development agencies, and media outlets would disseminate evidence-based guidance on scam recognition and protective behaviors.

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Appendix

The survey instrument and implementation code used in this study are available at the following locations:

A. Survey Form

The anonymous job scams survey administered to participants can be accessed at:
https://docs.google.com/forms/d/e/1FAIpQLSdzFKOwfvce9N-k-hZjGvT5LQyNGX7ufN_VILyp7ZZ_IBPmWQ/viewform

B. Survey Implementation Script

The Google Apps Script code implementing the survey logic, branching, and data collection is available at:
https://script.google.com/d/1mrGvZI9-NQNP_wqRuQfY4PKJB9PAXq9cAafELmFhg6IqwzP6DIvUNngs/edit?usp=sharing

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